Supplementary Material
Advice May Only Tighten: Untrusted Inputs to a Fail-Closed Vision-Language Gate on a Simulated Relief Robot

Anonymous Authors

Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

This document holds material moved out of the main paper. Section, table and figure numbers with an S prefix refer to this document; unprefixed references are to the main paper. All numbers are generated from the same result files as the main paper. It is organised as follows. *Methods and background:* S1 development data, T1 construction and the two-model rule; S2 model selection on development data; S3 the extended related work. *Results:* S4 additional T1 analyses (decision-rule ablations, risk–coverage, weaker models); S5 campaign E details; S6 wall-clock timing; S7 further result details (studies Z, P, R, campaigns F, G and H, hazard fixes). *Failure analysis:* S8 lessons from running the real stack. *Audit:* S9 the registration audit and threats to validity. *Moved from the main paper:* S10 untrusted inputs (proof of Proposition 2, exposure and steering, cost bound, trust base, consistency–robustness curve); S11 the second checker and campaign J; S12 all registered studies, the hazard-fix candidates with T2 and campaign F, study W, camera degradations (R), the zero-shot baselines (Z) and the model of repeated missions.

S1. Development Data, T1 Construction and the Two-Model Rule

Development data. Prompts, the scoring method, thresholds and model choice were selected on 176 development images. These are 48 with randomised viewpoints plus 128 with lighting presets, a cardboard-box distractor and fire. The instruction parser was tuned on 40 development utterances. The later hazard fixes (main paper, Sec. III-B) also used two hazard development sets of 300 rendered images each: DH, and DH2, which emphasises small early-stage fires.

Two-model rule. The two-model rule (Sec. S11) approves only if Eq. (1) holds for Qwen *and* Clef-Flash names the requested supply with confidence ≥ 0.65, presence ≥ 0.80 and hazard score below its own threshold.

T1 examples. Fig. S1 shows held-out T1 images. T1 used 60 Gazebo worlds $\times$ 2 stations $\times$ 5 viewpoints with continuous random lighting, supplies moved by up to ± 0.15 m and freely rotated, random fire positions, and three distractors. Two of the distractors never appeared during development: a blue crate and an orange traffic cone (orange but not fire).

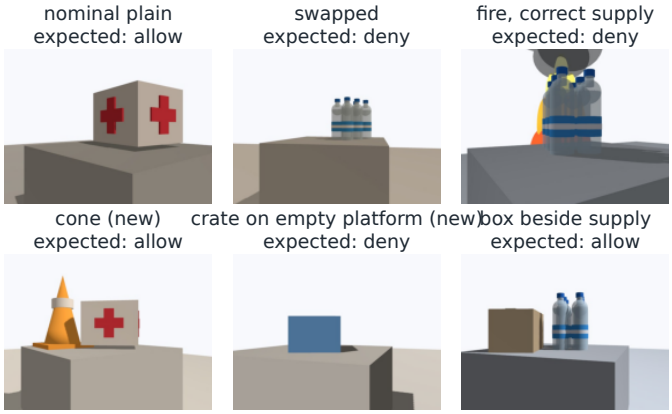


Fig. S1. Held-out T1 images (robot camera, 640×480) with the generator's expected decision. The crate and the traffic cone never appeared during development.

S2. Model Selection on Development Data

TABLE S1
CANDIDATE MODELS WITH THE FINAL PROMPTS (`v35`: CHECK QUESTIONS OF V3.5; `i35`: INSTRUCTION PARSER) ON DEVELOPMENT DATA ONLY. IMAGES: CORRECT DECISIONS WITH FALSE APPROVALS (FA) AND FALSE DENIALS (FD). EXPLANATIONS: ONE-SENTENCE REPORTS JUDGED CORRECT AGAINST THE GENERATOR LABELS. INTENT: DEVELOPMENT UTTERANCES, WITH UNSAFE ACCEPTANCES. VRAM IN MIB.

Model	Released	Images (176)	Explanations (128)	Intent dev (40)	VRAM
Qwen2.5-VL-3B	2025-01	176/176 (FA 0/FD 0)	84/128	31/40, unsafe 0	–
Qwen3-VL-8B	2025-10	176/176 (FA 0/FD 0)	120/128	38/40, unsafe 2	17212
Qwen3.5-4B	2026-02	175/176 (FA 1/FD 0)	125/128	35/40, unsafe 2	9102
Qwen3.5-9B	2026-02	176/176 (FA 0/FD 0)	125/128	35/40, unsafe 0	18320

Models were ranked lexicographically. The first key was zero false approvals and zero unsafe intent acceptances, and the second was explanation and intent accuracy. Qwen3-VL-8B parsed more development utterances but accepted two unsafe ones. Qwen3.5-4B produced a false approval. Qwen3.5-9B was therefore chosen and frozen before the T1 and E registrations.

S3. Extended Related Work

The main paper condenses this section; the full text follows.

Failure detection and verification with foundation models. REFLECT [1] and AHA [2] explain manipulation failures in natural language after the fact. Inner Monologue [3] feeds success detection back into a planner. Du et al. [4] fine-tune a VLM as a success detector. Sentinel [5], FAIL-Detect [6] and FIPER [7] monitor learned policies at run time (FIPER calibrates its out-of-distribution and uncertainty scores conformally on successful rollouts and requires both signals to alarm); Code-as-Monitor [8] checks task constraints with VLM-generated programs, both reactively and proactively, without a calibrated threshold or a ground-truth audit; an LLM-based anomaly monitor triggers reactive fallbacks [9]; and uncertainty-aware policy steering decides when to act, ask or learn [10]. *Closest prior art.* VAP-TAMP [11] (arXiv preprint, April 2026; we read the full text) verifies the preconditions and effects of each PDDL action with a VLM. For each predicate it generates several semantically equivalent yes/no questions, takes a majority vote and, if the answers agree, asks the VLM whether the view is sufficient; otherwise the VLM proposes a viewing direction and the robot moves there, up to a viewpoint budget. A failed verification corrects the scene graph and triggers PDDL replanning. It was evaluated by task success on 100 real mobile-manipulation trials and on injected failures in OmniGibson. The overlap with our design is real: paraphrase agreement is the counterpart of our permuted-question agreement, and VLM-chosen views are the counterpart of our look-around. To our knowledge, the differences are (i) our scores are next-token scores over a closed answer set with a threshold calibrated on development data, whereas VAP-TAMP uses textual yes/no answers and a vote; (ii) in our design every check is combined in a fail-closed conjunction, so a failed check or an extra view can only refuse or revoke, whereas in VAP-TAMP a verification result corrects the symbolic state in either direction; (iii) ours has an independent velocity guard that the VLM cannot command; (iv) in ours, memory may only reorder checks and tighten thresholds; and (v) ours is audited in pre-registered closed loop against simulator ground truth, which reports the unsafe-delivery rate and not task success. VAP-TAMP covers things we do not: PDDL task and motion planning, manipulation, and real-robot trials. Semantic anomaly detection with LLMs [12] and visual question answering that abstains rather than answers wrongly [13] are close in spirit to a gate that prefers refusal. Our check runs *before* the robot acts and is closed-set and zero-shot. Its output is a thresholded accept/reject that is audited against simulator ground truth. The closest methodological relative is KnowNo [14], which scores multiple-choice answers by next-token probability and asks a human for help, with a conformal-prediction coverage guarantee (see [15] for background on conformal prediction). Introspective planning [16] extends this line by letting the planner reason about task ambiguity. Conformal risk control [17] controls the expected value of any monotone loss (for example a false-negative rate), and Learn then Test [18] gives high-probability risk control; our study K (held-out fire miss rate $\leq \alpha$ with probability 0.9) is of that form. Conformal decision theory [19] calibrates a robot decision rather than a prediction set. Our gate is a one-shot accept/refuse threshold and not an online-adaptive controller. SayCan [20] likewise scores a fixed candidate set by language-model likelihood. We share the scoring, but we respond to uncertainty by refusing or searching, and we add permuted-option agreement as a test-time defence against letter bias.

Multiple-choice scoring and its biases. Scoring answer symbols works well when a model binds symbols to options [21]. Language models are reasonably calibrated on well-formatted multiple-choice questions [22], but they prefer certain option IDs [23] and orders [24], and their raw confidence needs validation [25], [26]. On embodied question answering, even the best frontier model reaches only 42.79% recall when it should abstain [27], which supports a calibrated refuse option over trusting the model to abstain. Our two permuted questions are a minimal, deterministic form of self-consistency [28]. The Yes/No hazard question follows the polling style of POPE [29].

Safety architectures. Simplex [30], [31], shields [32] and control barrier functions [33] keep a simple verified layer between a complex decision maker and the actuators. LLM-controlled robots can be manipulated into harmful actions [34], and guardrails such as RoboGuard [35] check plans before execution. The safety filter [36] is the current umbrella for such least-restrictive overrides of a nominal controller; our velocity guard is a minimal instance and the VLM check is a perception-side gate in the same spirit. Closest in using a VLM for safety, Santos et al. [37] turn images and language feedback into a failure set that a Hamilton–Jacobi safety filter enforces, so the guarantee comes from the verified filter; ours is a union bound over check errors. RoboSafe [38] guards VLM-driven embodied agents with executable safety rules and a short- and long-term safety memory, which drives rule-based replanning and is not constrained to tighten only. In contrast to vision-language-action models [39], [40], our VLM never commands motion.

Memory for robots in changing environments. Robots increasingly keep long-horizon semantic memory. ReMEmbR [41] and Embodied-RAG [42] retrieve from a database of captioned observations; KARMA [43] adds long- and short-term memory to embodied agents, and H-EMV [44] verbalises a robot's episodic memory. 3D scene graphs such as Hydra [45], ConceptGraphs [46] and HOV-SG [47] support planning [48] and language grounding [49]. Long-term mapping has long dealt with change [50], [51], and recent work tracks changes in dynamic scene graphs [52], [53]. These systems use memory to *answer* or *plan*. A recent preprint reports, in a small grid-navigation testbed, that acting on unaudited stale memory can be worse than having no memory [54]. Long-term agent memory is also an attack surface: corrupted entries can steer an LLM agent [55] (an LLM-agent result, not a robotics one). Memory benchmarks such as FindingDory [56] measure how well memory supports recall in long-horizon tasks, not whether acting on it is safe once the world has changed. GLOVE [57] and VLM-DEWM [58] check memory or a world model against fresh observations to

keep the memory *correct*.

Object search with priors and semi-static maps. Using past observations to decide where to look first is a classic idea, and we do not claim it as ours. Active visual object search keeps a belief over target locations and plans where to look next, using uncertain spatial and semantic knowledge [59], inter-object relations in Semantic Linking Maps [60], or object-oriented POMDPs whose belief factorises over objects [61]. Long-term mapping models objects or features that persist, appear or disappear (semi-static) and detects change [62], [63]; DovSG updates an open-vocabulary 3D scene graph locally as objects are moved [64]. Zero-shot object navigation with language or vision-language priors, such as ESC [65] and VLFM [66], uses a pre-trained model as a prior over where to explore, but has no mission-to-mission memory of a specific site. To our knowledge, these works optimise search efficiency (path length, success rate, map accuracy) and treat a detection, or an updated map, as the basis for the next action. We did not find in them a bound on the unsafe-action risk caused by *stale* memory at a pre-action gate. Our memory is narrower and serves a different goal: it may only reorder stations and add caution, the on-site verification is never skipped or relaxed, and this restriction is enforced in code and checked by the run audit, which gives a union bound that does not depend on how old the memory is. These methods could supply the ordering prior in our setting, for example a belief over supply location, provided the gate stays unchanged; combining them is future work, and we have not compared search efficiency against them. We study a different use: memory decides *where and how carefully to verify*. We restrict it so that it can never reduce verification, and we state this as a provable invariant and test it under scripted change.

Active perception. Moving the sensor to resolve ambiguity is a classic idea [67], [68], and view planning for inspection is well studied [69]. Recent embodied agents explore until a VLM is confident [70] or let a VLM choose views [71]. Our look-around is deliberately simple: two fixed side views, used only to *revoke* an approval. This makes its effect auditable and its safety direction obvious.

Robustness and the sim-to-real gap. Image classifiers degrade under common corruptions [72], and so do detectors [73], adapted vision-language models [74] and video-language models [75]. Sim-to-real transfer is a recognised challenge in robotics [76]. Without a physical platform, we measure one part of this gap, sensor degradation, under a pre-registered protocol. What we care about is the *direction* of the errors, not only accuracy.

Untrusted inputs to embodied agents. Jailbreaks of LLM-controlled robots are now well documented: RoboPAIR [34], BadRobot [77], the policy-executable attacks of Lu et al. [78] and adversarial text attacks on vision-language-action models [79] all steer the planner through the instruction channel. A second line puts the payload in the scene: signs and printed text in the camera view hijack VLM-controlled robots [80], [81], and sensory channels (OCR, audio, LiDAR) inject prompts into ROS 2 robots [82]; typographic vulnerability of large vision-language models [83] and indirect prompt injection in LLM-integrated applications [84] are the underlying mechanisms. Memory is a third channel: poisoned or injected memory entries steer LLM agents [55], [85], and a recent survey lists long-term memory and multi-agent communication as under-explored attack surfaces of embodied agents [86]. Defences in robotics are mostly filters or certification of the plan: RoboGuard grounds rules in a root-of-trust LLM and checks plans against them [35], and the mitigations proposed alongside the attacks are prompt defences, two-stage verification and text masking [78], [81]. The closest conceptual relatives are structural defences for software agents: CaMeL separates control and data flow and attaches capabilities to tool calls [87], and information-flow control labels the integrity of inputs and enforces policy deterministically [88]. We carry this idea to a physical pre-action gate: advice from memory, peer robots, inventory and operator text is typed so that it can only reorder visits or tighten thresholds, never skip or relax a check, which yields a bound on unsafe deliveries that holds for any advice (the tighten-only rule) and is evaluated closed-loop against poisoned memory and injected instructions (study A). Text in the scene is perception, not advice, and so lies outside this guarantee; we measure it separately (study S) and report the residual vulnerability.

Diverse redundancy. Requiring agreement between independently developed versions is a classic fault-tolerance technique [89]. Its benefit is limited by correlated failures [90], and model-agreement schemes for language models have reappeared recently [91]. We report both sides: the two models agree on most images, and the second model's value depends strongly on how it is asked.

Evaluation practice. We follow recommendations to report conditions, multiple metrics, uncertainty and failure modes [92]. We use Wilson intervals [93] and exact McNemar tests [94]. Domain randomisation [95] is used here to build a held-out *test* distribution, not for training. Our design is fixed-n and pre-registered; sequential testing with valid early stopping [96] is a possible refinement for small n. How far simulation predicts real-world behaviour is itself measurable [97]; we make no claim beyond simulation.

Fire and smoke perception. We use D-Fire [98] for study P because it is CC0 and ground-level with negative images; FASDD [99] is a larger alternative with ground-level, UAV and satellite subsets and a possible future source. Local open-weight VLMs have been used for wildfire detection from UAV imagery [100], descriptively and without refusal semantics or a closed-loop audit.

S4. Additional T1 Analyses

Free-form generation. Free JSON (A5) gave no parse errors with Qwen3.5-9B and the same 5 false approvals, but only 158 correct approvals against 187 with scoring. All models missed largely the same small fires (hazard recall 59 to 68 of the 76 visible fires); Supp. Fig. S2 shows descriptive risk–coverage curves over τ_s.

TABLE S2
T1 HELD-OUT TEST, REGISTERED V3.4 RULE (EQ. 1) UNLESS MARKED FREE JSON; 408 MUST-REFUSE AND 191 MUST-APPROVE IMAGES.

Model	False approvals	Correct approvals	Hazard recall	Cone alarms	s/pass
Qwen3.5-9B	5/408	187/191	67/76	0/25	0.98
Qwen3.5-4B	9/408	190/191	59/76	0/25	1.06
Qwen3-VL-8B	4/408	183/191	68/76	0/25	0.46
Qwen2.5-VL-3B	4/408	187/191	68/76	0/25	0.83
Qwen3.5-9B (free JSON)	5/408	158/191	–	–	–
Qwen2.5-VL-3B (free JSON)	0/408	57/191	–	–	–

TABLE S3
DECISION-RULE ABLATIONS ON T1: FALSE APPROVALS / FALSE DENIALS. *: EXACT MCNEMAR $p < 0.05$ VERSUS THE FULL RULE OF THE SAME MODEL.

Decision rule	Qwen3.5-9B	Qwen3.5-4B	Qwen3-VL-8B	Qwen2.5-VL-3B
Full rule	5 / 4	9 / 1	4 / 8	4 / 4
A1: question 1 only	5 / 4	9 / 0	4 / 7	4 / 4
A2: question 2 only	5 / 1	9 / 1	4 / 2*	36 / 1*
A3: no hazard question	47 / 4*	48 / 1*	46 / 8*	48 / 4*
A4: no threshold (τ_s=0)	5 / 4	9 / 1	4 / 8	4 / 4
v3.5: τ_h=0.05 (I-10)	3 / 4	4 / 1	4 / 8	1 / 30*

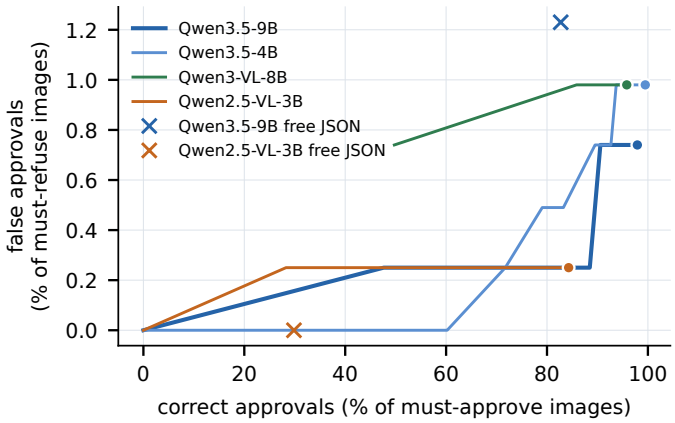


Fig. S2. T1 risk–coverage: false-approval rate versus correct-approval rate as τ_s varies ($\tau_h = 0.05$, v3.5). Markers show the frozen $\tau_s = 0.5$. The curves are flat over most of the range: the remaining errors are missed fires, which τ_s cannot fix.

A. Decision-rule ablations

Table S3 isolates the parts of the rule on T1 (Table S2, Eq. 1 of the main paper). For the 9B model neither the permuted second question nor the confidence threshold changed the number of false approvals. With Qwen2.5-VL-3B, requiring the two multiple-choice questions to agree reduced false approvals, which is the option-letter bias reported in [23].

B. Risk–coverage

Fig. S2 varies τ_s only for description, because τ_s was not tuned on T1.

C. Weaker models and free-form output

For Qwen2.5-VL-3B the second question alone approved 36 must-refuse images, and requiring agreement with the first question brought this down to 4, which is the option-letter bias reported in [23] in action (Table S3). Asking this model for free JSON (A5) left 148 of the 1200 generations unparseable and approved 57 images; Qwen3.5-9B gave no parse errors (main paper, Sec. VI).

S5. CAMPAIGN E DETAILS

Campaign E used a new randomised world in every trial (main paper, Sec. VI); Table S4 gives the outcomes by stratum and Fig. S3 the ground-truth trajectories.

S6. WALL-CLOCK TIMING

Where the time goes. Table S5 breaks down the 34 delivered missions that needed no search. The check itself takes about 5.5 s: aiming, waiting until stationary, frame checks, and the scored VLM queries (3.4 s). Driving between station and shelter takes about 39 s (sum of phase medians). The check is therefore short compared with driving. The situation report adds about 5 s but could run in parallel with the drive to the shelter.

TABLE S4
CAMPAIGN E (RANDOMISED WORLDS) BY STRATUM.

Expected	Condition	n	Correct	Observed
allow	box	2	2	allow 2
allow	cone	4	4	allow 4
allow	crate	2	2	allow 2
allow	plain	12	12	allow 12
deliver after search	box	2	2	allow after search 2
deliver after search	crate	2	2	allow after search 2
deliver after search	plain	12	12	allow after search 12
deny	box	2	2	deny 2
deny	crate	4	4	deny 4
deny	plain	6	6	deny 6
deny hazard	fire	8	8	deny hazard 8
deny hazard after search	fire other	4	4	deny hazard after search 4

S7. FURTHER RESULT DETAILS

Real photographs (studies P and P2; full text moved from the main paper). We also asked the frozen v3.6 hazard question about 400 real photographs from the test split of D-Fire [98], a CC0 fire and smoke dataset (100 each with no fire, smoke only, fire only, and both). It flagged 298/300 of the photographs with fire or smoke (95% CI [97.6, 99.8]%) and 64/100 of the fire-free ones ([54, 73]%); the false alarms concentrate on hard negatives such as hazy hillsides and sunset clouds (Supp. Sec. S7). The scores still rank the photographs well (AUROC 0.96, not registered), so the problem is the threshold, not the question. The question thus errs strongly toward stopping on real images, which is the safe direction but would cost availability outdoors, so a site-specific threshold calibration is a deployment prerequisite. Study P2 tested this: it chose wording and τ_h on 400 separate calibration photographs (recall ≥ 0.95 first, then the lower false-alarm rate), with wording `h35` and $\tau_h = 0.3$, and applied it to 400 test photographs. It flagged 282/300 of those with fire or smoke ([90.7, 96.2]%) and 18/100 of the fire-free ones ([11.7, 26.7]%), against 294/300 and 34/100 for the frozen $\tau_h = 0.05$. The registered claim (test recall ≥ 0.90 and false-alarm rate ≤ 0.20) was supported.

Conformal τ_h (study K, moved from the main paper).

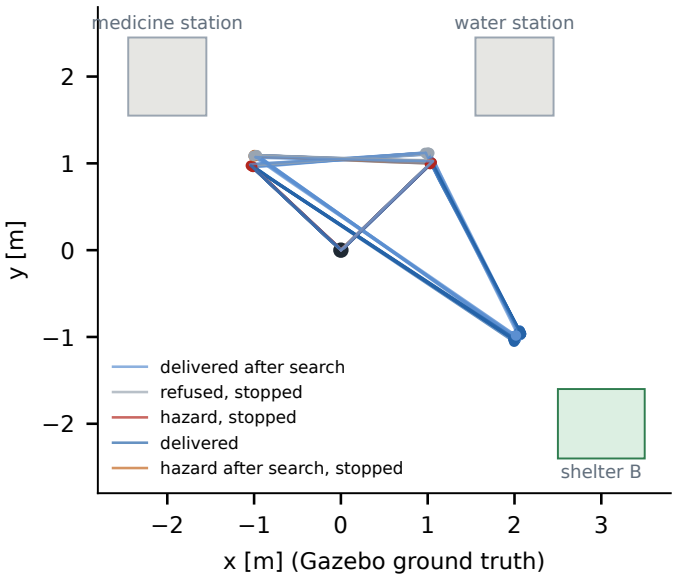


Fig. S3. Campaign E: ground-truth trajectories of all final trials, coloured by audited outcome.

TABLE S5
WALL-CLOCK PHASES OF DELIVERED MISSIONS WITHOUT SEARCH (CAMPAIGNS D AND E; V3.4 AND V3.5, NO LOOK-AROUND).

Phase	Median [s]	Min	Max
drive to station	14.31	12.19	16.62
aim camera	0.50	0.50	0.52
wait until stationary	0.67	0.66	0.71
acquire and check frames	0.90	0.75	0.98
VLM decision (scored queries)	3.44	2.95	6.64
situation report (not used to decide)	5.14	3.99	8.54
drive to shelter	24.83	15.02	25.32

Split-conformal calibration of τ_h on 211 development fires (miss rate $\leq \alpha$ with probability 0.9) gave $\tau_h = 0.0050$ ($\alpha = 0.05$) and 0.0113 ($\alpha = 0.10$). On the 76 visible T2 fires the bound held at $\alpha = 0.05$ and held at 0.10 (0/76 and 0/76 missed), with 26/193 and 4/193 false alarms on 193 fire-free must-approve images, against 0/76 and 0/193 for the frozen $\tau_h = 0.05$. Exchangeability is broken by design (small fires: 27% of calibration, 7% of T2 fires); the frozen threshold was not changed.

Instruction parsing (moved from the main paper). On 60 held-out utterances the hybrid parser was correct for 55/60 with 0 unsafe acceptances (TF-IDF parser: 31/60, 2 unsafe); all five errors were refusals, so they fail closed.

Study Z. The contrastive baselines used SigLIP [101] and CLIP [102] prompts for the supply and the hazard, and the same crops as our check. On development data no threshold pair gave both few false approvals and many correct ones. The contrastive scores do not separate "right supply, no fire" from the alternatives in these scenes.

T1 hazard-question false alarms. With the registered v3.4 rule, the hazard question raised false alarms on 0/25 traffic-cone images and on 0/523 images without fire. With the v3.5 rule, it raised false alarms on 0/25 cone and 0/523 fire-free images.

TABLE S6
STUDY Z: ZERO-SHOT CONTRASTIVE BASELINES VERSUS OUR CHECK ON THE HELD-OUT SETS (FALSE APPROVALS / CORRECT APPROVALS).

System	T1 FA	T1 CA	T2 FA	T2 CA
Ours (T1 v3.5, T2 v3.6)	3/408 (0.7%) [0.2, 2.1]	187/191 (97.9%) [94.7, 99.2]	0/407 (0.0%) [0.0, 0.9]	190/193 (98.4%) [95.5, 99.5]
SigLIP (so400m, zero-shot)	44/408 (10.8%) [8.1, 14.2]	191/191 (100.0%) [98.0, 100.0]	47/407 (11.5%) [8.8, 15.0]	193/193 (100.0%) [98.0, 100.0]
CLIP (ViT-L/14, zero-shot)	0/408 (0.0%) [0.0, 0.9]	2/191 (1.0%) [0.3, 3.7]	0/407 (0.0%) [0.0, 0.9]	0/193 (0.0%) [0.0, 1.9]

TABLE S7
STUDY R: V3.6 OFFLINE CHECK UNDER CAMERA DEGRADATIONS (T2 SUBSET). SEVERITY M = MODERATE, S = SEVERE.

Degradation	Sev.	False appr.	Fire missed	Correct appr.	Gate rej.
None (clean)	–	0/192	0/27	107/108	0
Sensor noise	M	0/192	0/27	107/108	0
Sensor noise	S	0/192	0/27	107/108	0
Motion blur	M	0/192	0/27	106/108	0
Motion blur	S	0/192	0/27	101/108	0
Defocus	M	0/192	0/27	108/108	0
Defocus	S	0/192	0/27	108/108	0
Under-exposure	M	0/192	0/27	107/108	0
Under-exposure	S	0/192	0/27	106/108	0
Over-exposure	M	0/192	0/27	96/108	4
Over-exposure	S	0/192	0/27	59/108	24
JPEG	M	0/192	0/27	108/108	0
JPEG	S	0/192	0/27	107/108	0

Campaign F, remaining misses. Six of v3.6's seven remaining misses were fires of 1 100–2 400 visible pixels with hazard scores of 0.002–0.006, which overlap those of fire-free views, so no threshold can separate them. The seventh showed 8 600 pixels at $p_h = 0.019$.

Campaign G, world set and medians. The set of worlds is defined from mask renders at all three views, so it includes the side views by construction. The medians over all delivered runs, 102 s and 122 s, are not comparable, because look-around refused most of the fire runs that the front-only system delivered.

Study P, false alarms and licence. D-Fire [98] was released by its authors under CC0, as stated in the original repository's LICENSE; we used a Hugging Face mirror. Rendered fire is stylised, which motivated this check on real photographs. D-Fire's fire-free class comes from fire-surveillance cameras and is rich in hard negatives: hazy hillsides, low clouds, a smudge on the lens, red sunset clouds, glowing tents. Everyday scenes such as streets, beaches and roads scored $p_h \approx 0.01$. The question was worded to catch faint smoke ("a grey or black cloud") and thresholded at 0.05, so it errs strongly toward stopping on real images. We registered the site-specific threshold calibration as a follow-up study (P2).

Campaign H, failure types. Staleness predicts two failure types for naive memory: delivering a supply that has been moved away, and delivering where a fire has appeared since the last visit; the per-trial ledger records which occurred.

Study R, all conditions. Table S7 gives the false approvals, correct approvals and acquisition-gate rejections of study R (Sec. S12-E) for every degradation and severity.

Hazard fixes on development data. Table S8 compares the candidate hazard fixes of the main paper (Sec. III-B).



Fig. S4. Study R: one held-out frame with a visible fire under the six registered degradations (severe level; the moderate level is milder). Over-exposure washes out the flames in this example. Frames that the acquisition gate rejects count as refusals.

TABLE S8
HAZARD FIXES ON DEVELOPMENT DATA (DEV V1+V2, DH, DH2). $\tau_h = 0.05$ UNLESS STATED.

Configuration	False appr.	Correct appr.	Fires refused	One V100
v3.5: Qwen3.5-9B	46/413	357/360	165/211	yes
+ enlarged crops	37/413	356/360	174/211	yes
+ hidden-fire wording	25/413	357/360	186/211	yes
+ wording + crops (v3.6)	17/413	357/360	194/211	yes
Qwen3.5-4B	37/413	360/360	174/211	yes
Clef-Flash, original Jev request	45/413	182/360	166/211	yes
Clef-Flash, task wording	42/413	350/360	169/211	yes
Clef (task) + Qwen (v3.6), both approve	15/413	350/360	196/211	no

TABLE S9
SECOND-MODEL OVERLAP ON THE 211 VISIBLE FIRES OF THE DEVELOPMENT SETS DH AND DH2 (NO TEST DATA). A FIRE IS MISSED WHEN THE CHECKER WOULD APPROVE THE VIEW (QWEN3.5-9B V3.6, $\tau_h = 0.05$; CLEF-FLASH WITH TASK WORDING, HAZARD THRESHOLD 0.05; SAME RULES AS TABLE S8). SCRIPT: `SCRIPTS/JEV_OVERLAP.PY`.

	Clef-Flash misses	Clef-Flash catches
Qwen misses	15	2
Qwen catches	27	167

Qwen misses 17, Clef misses 42; P(Clef misses | Qwen misses) = 88% (Wilson [66, 97]%); Jaccard 0.34. Counting only the hazard question (score ≥ 0.05 in the hazard question alone): Qwen misses 23, Clef misses 55, both miss 23.

Campaign H, per-mission outcomes. Fig. S5 shows every mission of campaign H (main paper, Sec. VI-B).

Campaign H, case study. Fig. 5 of the main paper shows the one mission in which M0 delivered into a fire (sequence Q09, mission 3).

Memory caution versus a global threshold (descriptive, not registered). Campaign H stored decisions, not every view's p_h, so only a partial comparison is possible from existing logs. *Fires:* the one mission where caution changed an outcome is Q09.3 (M0 $p_h = 0.035$, M2 0.049); a global $\tau_h = 0.0113$ (study K, $\alpha = 0.10$) would have stopped M0 there too, since $0.035 > 0.0113$. *False stops:* on the 193 fire-free T2 views the global $\tau_h = 0.0113$ gave 4/193 false alarms (2.1% per view, about 6% per approval with three views if independent), against 0/193 at the deployed 0.05. Caution applied in 12 of the 32 M2 missions: 8 at stations with a fire (all 8 stopped) and 4 at fire-free stations, with 0 false refusals there (Wilson upper bound 49%). That is compatible with both zero and the global rate, so a comparison at equal false-stop rate cannot be made from these logs. What caution does buy by construction is lower exposure: the strict threshold applies at stations with a recent hazard, not everywhere. A direct test needs many fire-free missions at stations with a hazard history; we did not run one.

A. *Campaign H: all paired missions*

Table S8 lists every paired mission of campaign H (missions 2 and 3 of the 16 sequences) with its outcome and mission time under M0, M1 and M2.

Repeated-mission model. The model of repeated missions, with the per-check rates measured in G and H ($p_{\mathrm{det}} = 0.95$, $n = 44$), is shown in Fig. S8 (Sec. S12-F); its curves are computed from measured rates, so their order is a model prediction, not an observation. The figure also shows a fourth policy, M1age, a naive memory that is trusted only while its entry is recent (recency-gated, without verification); it lies between M1 and M2 in unsafe deliveries. Over the whole range of change probabilities $q \in [0, 0.5]$ the model finds no break-even point: M1 and M1age stay faster than M0, and M2 is faster than M0 at small q and converges to it as q grows, because M2 re-checks and so approaches the no-memory behaviour. At $q = 0.2$ per mission the unsafe deliveries per 100 missions are 35.83 for M1, 1.95 for M2 and 1.95 for M0, with mean mission times 32.4, 71.7 and 80.3 s. Neither the single missed-fire delivery nor the 32 paired missions measure how often a real world changes; q is a free parameter of the model.

More stations (descriptive, not registered). Our arena has $k = 2$ stations, so memory can save at most one search detour. Let the supply sit at one of k stations and move, with probability q between missions, to a uniformly chosen other station, so its long-run location is uniform. Without memory the robot visits stations in inventory order and needs $(k+1)/2$ visits on average; with verified memory it first visits the station where it last saw the supply and needs $1 + qk/2$. With the detour $\delta = 30$ s measured in H, the expected saving is $\delta\,((k-1)-qk)/2$ per mission and vanishes at $q^* = (k-1)/k$. For $k = 2$ this reproduces the Monte-Carlo model above (9 s at $q = 0.2$); for $k = 8$ it is 80 s. A simulation of the same process agrees within 1 s for $q > 0$ (`scripts/h_k_scaling.py`). The safety side does not depend on k in this way: each policy checks every station it delivers from with the same rule, so Proposition 2 applies unchanged, and memory lowers the expected number of checks per mission rather than their strength. The model assumes the stations are equally far apart and that supply identification is perfect (T1: no supply false approvals in 332 images).

S8. FAILURE ANALYSIS AND LESSONS

Running the full stack on a GPU revealed 28 problems that the 1253 offline test functions and mocked ROS runs had not; the most instructive are listed here (the full table is in `docs/FINDINGS.md` in the repository).

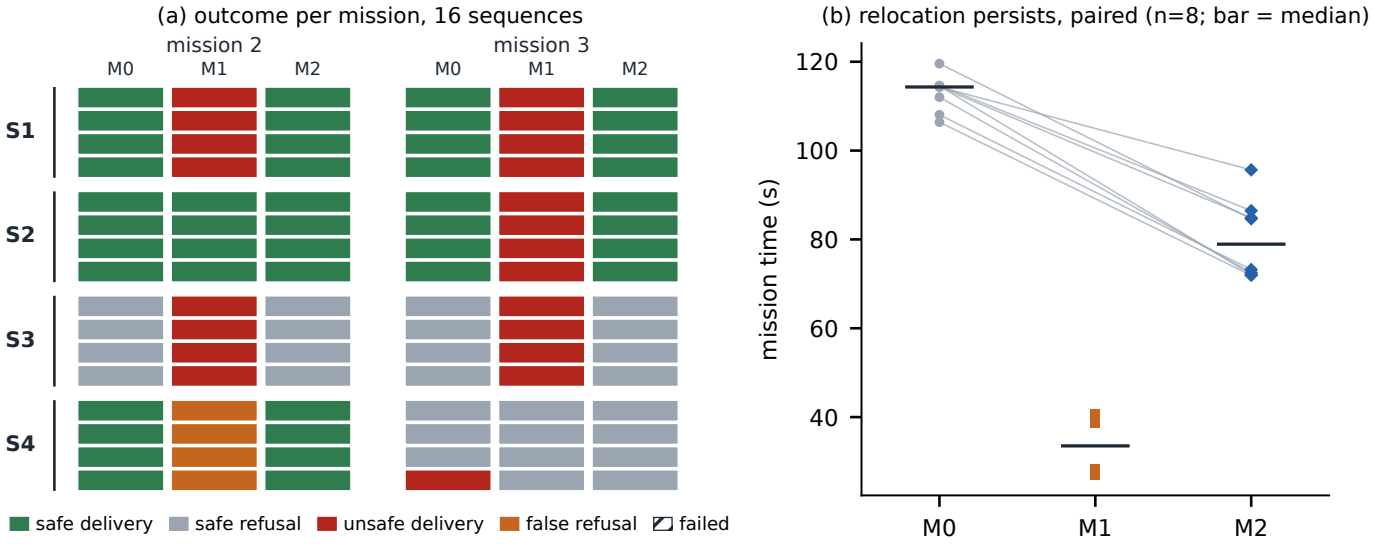


Fig. S5. Campaign H, missions 2 and 3 of the 16 sequences. (a) Outcome of each mission under no memory (M0), naive memory (M1) and verified memory (M2); rows are sequences, grouped by scripted story (S1–S4; definitions in the caption of Table S8). No mission failed (the hatched "failed" category is empty). (b) Mission time of the eight missions whose relocation persisted, for each arm; grey lines join the M0 and M2 times of the same mission, bars mark medians. M1 times are short because M1 skips the checks.

- **The auditor trusted the controller's sensor.** The wheel joint axis was expressed in the child-link frame after a 90° roll, so the wheels spun about the vertical axis. Odometry reported 1.5 m forward motion while the body slid sideways by 0.75 m. An audit based on `/odom` would have passed the run. Audits now use Gazebo's ground-truth odometry. Changing the wheel collision from a cylinder to a sphere reduced dead-reckoning drift on the medicine route from 0.16 to 0.011 m (worst of two repeats; `scripts/i14_pilot_p_range.py`).
- **Free-form JSON from the first VLM was unusable** on our renders. With Qwen2.5-VL-3B, 21 of 96 responses (48 images, two questions each; all 21 for the first question) could not be parsed and the model hallucinated water on empty shelves. A newer model parses reliably, but it still approves fewer correct deliveries than scoring does (Table S2). This motivated closed-set scoring. T1 measured the free-JSON method again (Table S2).
- **Middleware effects.** Best-effort QoS silently dropped 92 of 102 0.9 MB camera frames under load. The GPU lidar detected the robot's own camera mast. Slow DDS discovery on a 4-core host failed 4 of 12 starts in an aborted campaign. An interrupted supervisor left an orphaned stack that made every trial of the next campaign stop with two `/cmd_vel` publishers. All of these are recorded with their fixes; the aborted campaigns are published.
- **Stale-request latch.** Two pre-decision retries in campaign D happened while the robot was stationary: the mission node's request stream stalled for ≈0.3 s on the saturated CPU. We instrumented the mission loop, the guard and garbage collection and rejected two hypotheses (guard starvation, GC pauses). Raising control-loop priority did not remove the stalls. We therefore changed the guard: a stale zero request now holds zero, while a stale motion request still latches. With a 0.6 s stall injected before departure, the old guard failed 2/2 runs and the new one completed 2/2. Raising priority with a file capability then had a side effect: it put the exec into the dynamic loader's secure mode, which stripped `LD_LIBRARY_PATH` and broke the ROS bag recorder. A regression run caught this before any campaign.
- **A pilot exposed a threshold that was not conservative enough.** Before campaign E we ran nine trials in separate pilot worlds. In one, the robot found the wrong item, searched the other station, and delivered from it although a fire burned behind it. The fire was almost entirely hidden by the first-aid kit, and $p_h = 0.32$. On development data no fire-free image exceeded $p_h = 0.0007$, so we lowered τ_h to 0.05 (v3.5). The same pilot worlds then gave 9/9 expected outcomes. T1 was not involved: both thresholds were registered for T1 before any T1 image was scored.
- **A generator bug caught by looking at the data.** In the first T1 render, distractors meant to stand next to a supply were placed through it. The emptiness check ran after the supplies were moved. The render was discarded and the fix recorded as a pre-registration amendment, before any model scored a T1 image. A second amendment replaced the field-of-view fire label with the rendered-mask label, after we saw fires hidden behind supplies that the geometric label had called visible.

S9. Registration Audit

For every study with results, the registration commit precedes the results commit (holds). Campaign C's registration and ledger entered the repository together, so their order

cannot be verified from the history. Studies whose text is marked "draft" (A, S, X, XD, Q, B, IF; study Q's design is still on its own branch) have no results yet: their registration is the commit that puts the frozen text on `main` (the merge of the integration branch), and no trial or inference of them is run before that. The "Registered" column shows the newest commit of the text in force at the first results (or of the draft, while there are none).

A. Threats to validity

Internal. The studies were designed sequentially, each registered after the results of the previous ones were known, so registration is confirmatory within a study but not across the programme (Table S9); study K was registered after the T2 scores it reuses had been computed (`PREREGISTRATION_K.md`, offline, no new inference). The H stories are scripted so that a naive memory fails by construction; the M1–M2 contrast is therefore large by design, and the M2–M0 comparison is the one that bounds risk (`PREREGISTRATION_H.md`). The analysis scripts computed the median of an even-length list as the upper-middle element; this was fixed and all summaries regenerated, and errata with no change to protocols, endpoints or claims are in `PREREGISTRATION_{F,G,H}.md` (for H, a relocation-persist median of 79 s had been reported as 85 s). Campaign J deviated during the run (`PREREGISTRATION_J.md`): from trial J007 documentation builds shared the 4-core host, and alternating about 46 GB of weights through a 54 GB page cache made every check re-read its weights, so measured latency includes model I/O and is inflated in both arms; two first attempts failed before any VLM decision and were re-run under the registered retry rule.

Construct. Fire labels come from rendered mask pixels (a fire is perceivable at 614 pixels or more), not from human judgement, and the fire is stylised. "Delivery" means that the robot reached the shelter after approving; there is no handover, so grasping and placing are not tested.

External. All results use one arena geometry, two stations and simulation only. The real-photograph study (P) is descriptive and reports false alarms on D-Fire images; it does not test a robot. The two checkers in J share lineage (Clef-Flash is post-trained from Qwen3.5-9B), so errors may be correlated.

Statistical. Samples are small: the time comparison has 8 relocation-persist pairs, and the safety comparison contrasts 0 unsafe deliveries (M2) with 1 (M0), which shows only that M2 is not less safe (Wilson upper bounds 11% and 16%). Images of one world, and missions of one sequence, are not independent; we therefore add world-cluster bootstrap intervals and a sequence-level McNemar test, which are wider or less powerful than the pooled ones. Several studies are reported without family-wise correction, because each has its own registered endpoint and the claims are not pooled.

S10. Untrusted Inputs: Proof, Exposure, Cost Bound and Trust Base

Setting. Let $A_\theta(o) = [\text{supply}(o;\tau_s)] \wedge \bigwedge_{v\in V}[p_h^{(v)}(o) < \tau_h]$ be the check of Eq. (1) with the look-around views V, and let an advice set $\mathbf{a} = \{a_1,\dots,a_n\}$ be any finite set of items $a = (\text{source}, \text{kind}, \text{station})$ with kind $\in \{\text{location}, \text{hazard}, \text{safe}, \text{skip}\}$ and station one of the site's k stations ($k = 2$ in the arena, 4 in the depot). The planner is the function $\text{plan}(\theta, \mathbf{a}) = (\pi, \theta')$ implemented in `rescuemind/advice.py`:

- π is an order of the k stations: the first applicable location item for station j moves j first (later ones, an item naming the catalog station, and any item after a reorder by memory are rejected and logged); all other items leave π unchanged. The set of stations is never changed, so π is a permutation of the order before advice.
- $\theta'_j = (\tau_s,\ \tau'_{h,j},\ V)$ at station j, with $\tau'_{h,j} = \min(\tau_h, \tau_h^{\text{caut}})$ if some hazard item names j and $\tau'_{h,j} = \tau_h$ otherwise, where $\tau_h^{\text{caut}} = 0.01$.
- *safe* and *skip* items are rejected and logged; no item can create a refusal without a visit, a delivery without a check, or a change of τ_s or V.

Two optional hardenings, both off by default and both on the tighten-only side, change θ' further. *Scrutiny* gives the station that location advice moved to the front the caution threshold and the extra views of the adaptive rule (two poses at 0.8 m and one 0.3 m closer), whether or not anything looked suspicious; it only lowers τ'_h and enlarges V'. *Soft refusal* turns a refusal caused only by the caution threshold ($\tau_h^{\text{caut}} \le p_h < \tau_h$, nothing else wrong) into a refusal of that station and a continued search; a delivery still needs an approval of the full check at the delivering station, a real hazard ($p_h \ge \tau_h$) in a view already taken still stops the mission (the first caution-only side view ends the station's look, so a hazard in the unseen side view is neither acted on nor recorded; that station is never delivered from), and the audit accepts a soft hop only if the saved model outputs recompute it as caution-only.

Lemma 1 (every plan is verification-preserving). For every $\mathbf{a}$, $\tau'_{s,j} = \tau_s$, $\tau'_{h,j} \le \tau_h$ and $V'_j \supseteq V$ at every station j in π. *Proof.* Immediate from the definition of plan: the second component is a minimum with τ_h, the first is copied, the third is copied or enlarged. The run audit recomputes $\text{plan}(\theta, \mathbf{a})$ from the logged items and rejects the run if the logged θ' differs or is not of this form (`advice.assert_monotone`). □

The error budget. *Probability space.* A sample point is $\omega = (w, \xi)$: w is a world from the generator (supply arrangement, fire, lighting, props) and ξ is exogenous sensing and model noise (the server's answers are a deterministic function of the image, cached by its digest, so ξ is rendering and arrival-pose noise). For a station s and a pose p of the finite design set $\mathcal{P}$ (front, L, R and the three extra poses) the frame is $o_{s,p} = F(w, s, p, \xi_{s,p})$. For $\theta = (\tau_s, \tau_h, V)$ with $V \subseteq \mathcal{P}$ the approval is $A_\theta(o_s) = C(o_{s,\text{front}}; \tau_s, \tau_h) \wedge \bigwedge_{v\in V}[p_h(o_{s,v}) < \tau_h]$. $M_s(w)$ says that s is *must-refuse*: it holds the wrong or

no supply, or a fire perceivable (at least 614 pixels) at one of the three design views, evaluated at check time. *Unsafe* is defined as a delivery from some s with $M_s(w)$; a fire below the perceivability threshold in all three views is counted by the ground-truth audit, but reported separately as outside the guarantee. Counting it as unsafe would make the unsafe event larger than E below. *Assumptions.*

- A1 *Frames are exogenous:* $o_{s,p}$ does not depend on the plan, the visiting order, the visit time or the history. The pointwise step "$A_{\theta'}(o_s) = 1$ implies $A_\theta(o_s) = 1$ on the same frames" is a coupling that is only meaningful under A1; under advice the robot captures different realised frames than the baseline robot, so without A1 the implication says nothing about the law of o_s. The world is also static between the check and the delivery. A1 holds in the arena with a fixed arrival pose; real rendering and arrival-pose jitter satisfy it only approximately.
- A2 *The plan only tightens:* for every advice set $\mathbf{a}$ the plan gives each station s parameters $\theta'_s = (\tau_s, \tau'_h, V')$ with $\tau'_h \le \tau_h$ and $V' \supseteq V$ (Lemma 1, audited by `assert_monotone`), and the base plan is the baseline or verified memory. A naive-memory base (skipped checks) passes the plan invariant relative to itself but is outside the proposition; the interlock rejects any plan with a skipped check or a refusal without a visit.
- A3 *Gate:* the robot delivers from s only if $A_{\theta'_s}(o_s) = 1$ on the frames it captured at s. This is a property of the mission node, not of the planner: every path to the delivery leg (first station, recovery hops, soft refusal, k-station search) passes through an approval of the *current* station (checked in `gazebo_waypoint_mission.py`, `stations.validate_visits`, `caution_soft.audit_caution_refusals`); it is verified after the run by audit and before the delivery only if the interlock (I-22) is on. A broken view (an invalid or non-finite hazard score, an inference error) counts as a hazard, not as clear.
- A4 *Advice does not influence* (w, ξ) (a causal, not a probabilistic-independence, statement: the advice is itself a function of ω): the adversary may be any function of ω (clairvoyant: it sees all frames), adaptive across missions, but cannot change w, ξ, the generator's law, the model, the code or the guard. Scrutiny requires look-around; the audit rejects a scrutinised approval without the extra views.

Let $E = \bigcup_{s=1}^{k} \{M_s(w) \wedge A_\theta(o_s) = 1\}$, $\varepsilon_s = \Pr[M_s \wedge A_\theta(o_s) = 1]$ (over worlds and noise), $\varepsilon_{\max} = \max_s \varepsilon_s$ and $\varepsilon_{\text{any}} = \Pr[E]$. Then $\varepsilon_{\max} \le \varepsilon_{\text{any}} \le \min(1, \sum_s \varepsilon_s) \le k\,\varepsilon_{\max}$. ε_{any} depends on the site and weakly increases with the station set; it does not depend on the visiting order or on the advice. The numbers plugged in for a class (the hard one) are conditional on a class event $C(w)$, and the bound then reads $\Pr[U \mid C] \le \Pr[E \mid C]$.

Proposition 2 (advice stays inside the error budget). Under A1–A4, for every mission and every rule that picks the order and the advice as a function of ω, $\Pr_\omega[\text{unsafe delivery when the advice is } \mathbf{a}(\omega)] \le \varepsilon_{\text{any}} \le \sum_{s=1}^{k} \varepsilon_s \le k\,\varepsilon_{\max}$. Over m missions, $\Pr[\exists\,\text{unsafe}] \le \min(1, m\,\varepsilon_{\text{any}})$ by the union bound; if the frames are deterministic and the world persists, E is the same event in every mission, so there is no accumulation.

Proof. Fix ω and an advice set. By Lemma 1 (A2) the supply conjunct is the same predicate and $V \subseteq V'$, so the hazard conjuncts of A_θ are among those of $A_{\theta'_s}$ (for every realisation of the frames: the views actually taken under θ', adaptive ones included, contain those taken under θ), and each is at least as strong under $\tau'_h \le \tau_h$: $[p_h^{(v)} < \tau'_h] \Rightarrow [p_h^{(v)} < \tau_h]$. A conjunction that contains a stronger version of each conjunct of another implies it, so $A_{\theta'_s}(o) \Rightarrow A_\theta(o)$ for every o. (The code does not compute a minimum: the caution is an extra conjunct $[\max(p_1, p_2) < 0.01]$ next to the server's label $\text{scene} \neq \text{hazard}$, and their conjunction equals $[p < \min(\tau_h, 0.01)]$; the server rounds p_h to four decimals, which can only make the caution stricter.) If the robot delivers from s and $M_s(w)$ holds, then $A_{\theta'_s}(o_s) = 1$ by A3, hence $A_\theta(o_s) = 1$ on the same frames by A1, so $\omega \in E$. This holds for every ω and for any order and advice that are functions of ω (A4), including a clairvoyant attacker. Taking probabilities gives ε_{any}, and Boole's inequality gives $\sum_s \varepsilon_s$, with no independence assumption (the sum counts stations that a mission never checks, so it is not tight). □

Sharpness. If the adversary knows ω and may name any non-default station first, taking $j^*(\omega) \in \{s : M_s \wedge A_\theta(o_s) = 1\}$ makes the first check at j^* approve, so $\Pr[\text{unsafe}] = \varepsilon_{\text{any}}$ is attained; the code allows this (any non-catalog station can be named first). A duplicate supply is needed only for the *control* to have risk 0, so that advice raises the risk; it is not needed for equality. *Without A1* (depot: the arrival pose at a bay depends on the previous bay and the smoke emitter moves, so detours change visit times) replace $A_\theta(o_s)$ in E by $\exists h \in H_s : A_\theta(o_s^h)$ over the reachable histories H_s; an average over histories, such as the stationary value that we measure, estimates a lower bound of this supremum and is not a bound, because the attacker chooses the order and hence the arrival history. Image-level estimates of ε_s (studies T1, W, XD) hold at the design poses; the closed-loop rate differs (T2, F).

Necessity, and what relaxing costs. Call an advice interface *relaxing* on a set R of (advice, frame) pairs if $A_{\theta'(\mathbf{a})}(o) = 1$ but $A_\theta(o) = 0$ there (a skip, a looser threshold, a dropped view, a trusted "already checked"). Under A1, A3 and A4 the proof above gives, for any interface, $\Pr[\text{unsafe}] \le \varepsilon_{\text{any}} + \varepsilon_{\text{rel}}$, where $\varepsilon_{\text{rel}} = \Pr[\exists s : M_s \wedge (\mathbf{a}, o_s) \in R]$ is the mass of must-refuse stations whose frames the advice can relax; for tighten-only interfaces $R = \emptyset$. The converse needs two premises. (N1) *No certificate of safety*: the relaxed frame o^* used below can be produced at a must-refuse station by some world of the generator family. A sound certificate, such as a verified sensor reading, would make a relaxing interface safe, and N1 excludes it. (N2) An approval leads to a delivery. Under N1 and N2, for every relaxing interface there is a generator law and a

clairvoyant advisor with $\Pr[\text{unsafe}] > \varepsilon_{\text{any}}$. Take $(\mathbf{a}^*, o^*) \in R$ and a law that gives mass q to a world whose must-refuse station produces o^*, and mass $1-q$ to worlds with check error ε_0. Send $\mathbf{a}^*$ exactly when o^* is captured. Then $\Pr[\text{unsafe}] \geq q + (1-q)\varepsilon_0'$, while $\varepsilon_{\text{any}} = (1-q)\varepsilon_0$, so any $q > 0$ suffices when the advisor also reproduces the remaining errors ($\varepsilon_0' = \varepsilon_0$); $q > \varepsilon_0/(1+\varepsilon_0)$ suffices in general. Under N1 and N2, the bound of Prop. 2 for every advisor and every law therefore fails for every relaxing interface; it holds for a relaxing interface only if its relaxed frames carry a certificate (outside N1), and the decomposition above is the quantitative form. The naive and filtered baselines are relaxing by construction (they skip the check on "safe"/"skip"), so no such bound exists for them, whatever their measured rate in study A. A filter shrinks R; it does not empty it.

Relative, not absolute. The proposition bounds the robot with advice, and the advice-free robot, by the same budget; it does not say that the two have the same risk. The advice-free robot's order is the inventory's, which is itself advice, and it checks the stations in that order and cannot be fooled by a station it never visits: the unsafe rate with advice can exceed the control's by up to the budget of the extra must-refuse stations that advice makes the robot check. We therefore do not claim "as safe as without advice". With $k = 2$ and the hardest class (a small fire, front pixels 0, side pixels 2 000–5 000), the plug-in is $\hat{\varepsilon} = 0.062$ with look-around (Wilson upper bound 0.28), so $k\hat{\varepsilon} = 0.125$ (upper bound 0.57); front-only it was 11/16. At $k = 4$ the upper bound exceeds 1, so as an absolute number the guarantee is vacuous for this class. The measured part of the budget is the point: it shows that ε, not advice, is what to reduce (Sec. III-B).

Exposure and steering. Advice that changes *which* station is checked changes exposure, in two ways. (a) *Delivery risk.* With duplicated supply (two bays hold the requested supply; $k = 4$ in the depot) location advice can put the bay with the hardest-to-see fire first, so the unsafe rate is $\varepsilon_{\text{hard}}$ where the control's is 0 by construction; truthful advice cannot be flagged by a filter, a provenance rule or a reputation system. In the 2-station arena only $\varepsilon_{\text{supply}}$ can be exploited (0/332 observed false supply approvals), which is why study A's first 162 runs cannot see this effect. The A-steer condition of study A builds the case: 16 duplicate-supply worlds in which a small, partly hidden fire (front view mostly 0 pixels, a side view a few thousand) stands behind the *other* station, truthful location advice naming it, with arms control (clean station first), tighten-only, scrutiny with soft refusal, and naive. The registered prediction is that tighten-only advice is worse than the control (about $\hat{\varepsilon}_{\text{hard}}$ unsafe worlds in 16) and scrutiny recovers part of it at a measured cost. Sixteen worlds cannot detect an effect of that size (about one unsafe world expected; a one-sided sign test needs five): a null is reported as undetectable at this n with its Wilson upper bound, never as evidence that the order does not matter. The steering estimate is the order-independent ε_{any}, measured at image level in studies W and XD (oracle steering, about 10 times the worlds). (b) *Physical exposure.* The robot is driven toward a station that really is on fire; the guarantee is about delivery, not travel. Both are reported as exposure (the share of runs whose first check was at a perceivable fire), never tolerated as noise.

What the proposition does not say. It does not bound time or availability; extraction of items from text may be wrong, which cannot matter for safety because the guarantee holds for *any* advice set; it assumes A1–A4 (a static world between check and delivery, and that ε_s is the measured per-station rate) (it differs by station and fire geometry); and it does not cover anything that reaches the decision through the camera. The destination in the operator's request is parsed separately in a closed set with an explicit rule, not by this proposition; a mis-parse of the *request* defeats everything, so the offline set of study A measures wrong-shelter acceptance (must be 0) as the evidence for it.

Scrutiny and soft refusal. Scrutiny lowers the risk of the steered station only; after a wrong-supply refusal there the search proceeds to unscrutinised stations, so the worst case is still ε_{any}, and its benefit is an empirical claim (A-steer), not part of Prop. 2. A soft refusal can continue a mission that the baseline would have stopped (a hazard seen later at that station); this is covered by ε_{any}, which is a union over all stations, not by the single-station argument.

Cost bound (denial of service), in time and availability. All bounds are per mission and against the robot without advice and with the same inventory order. Let W be the time of one extra station visit (a drive plus a front check; in study A the median mission time of the control in worlds where the supply is at the other station minus that in nominal worlds, taken from arms with the same look-around and adaptive flags), S the time of the three scrutiny views, W' the time of a soft-refused visit (drive, front check and side views), ℓ the value of a lost delivery, $\bar{\delta}$ the Wilson 95% upper limit of the per-view false-alarm rate of $p_h \geq \tau_c$ on fire-free views, *measured at the caution threshold* $\tau_c = 0.01$ ($4/193$ held-out fire-free T2 views, $\bar{\delta} = 5.2\%$; the count is the same at the conformal threshold 0.0113; `scripts/k_conformal.py`), and n_s the number of views actually taken at station s (3 under look-around, 6 under scrutiny: front, L, R, L_2, R_2, N). (i) *Reordering.* The plan applies at most one location item, so wrong or late location advice adds at most *one* extra station check relative to the no-advice order ($\leq W$; $k-1$ visits, $3W$ in the depot, is the regret against the *oracle* order, not the extra cost over the control); scrutiny adds S on every location-advised mission, truthful ones included. If the advised station really burns, the mission ends at the hazard stop without a delivery that the baseline would have made from the clean station: that is a lost delivery (an availability event with the probability that the advised station perceivably burns), not a delay of at most W. A truthful hazard report or location advice repeated at every mission is a time-budget attack of the same size per mission. (ii) *Caution.* A fire-free station that is checked under τ_c is falsely refused with probability at most $n_s\bar{\delta}$ (Boole): $3\bar{\delta} = 15.6\%$ with three views and $6\bar{\delta} = 31.3\%$ with six; if the views' alarms are positively associated (same scene, same look-alike object) the bound $1 - (1-\bar{\delta})^{n_s}$ is

14.8% and 27.5% (point estimates 6.1% and 11.8%). Only stations that are visited count: the second station of the arena is visited only after a wrong-supply refusal at the first, so the earlier worst case of $3k$ views is not attained. The views of one station are strongly correlated, which makes the union smaller; the per-station rate, clustered by station and world, is the quantity to measure. The rate was measured on held-out fire-free *images*, not on the closed-loop views of one station. (iii) *Price of a refusal.* Hard refusal: the delivery is lost. Soft refusal: at most W', and the delivery is lost only if no further candidate holds the supply; with a single copy of the supply, as in the arena, a soft refusal of that station still ends without a delivery, so the benefit of soft refusal is conditional on a duplicated supply (the depot). Hence $\mathbb{E}[\Delta t] \le W\,[\text{loc}] + S\,[\text{scr}] + \sum_{s\ \text{visited}} n_s \bar{\delta}\,(W'$ soft, or ℓ hard). In study A the sum of the per-run bounds over the 24 tighten-only runs on poison_location and poison_hazard worlds is 1.7 expected false refusals; `scripts/a_analysis.py` compares the observed count with the 95th percentile of the Poisson-binomial sum (4) and reports the extra seconds per attacked mission against $1.5W$. These are statements about the arena and the simulated rate, not about the rule: on real photographs the frozen $\tau_h = 0.05$ already gave 64/100 false alarms (P) and the calibrated threshold is 0.3 (P2), so the caution threshold should be defined relative to the site's calibrated threshold and its false-alarm rate re-measured per site. *Memory feedback.* The robot's own record is a second channel: before the I-19 fix, a caution-level alert (only the tightened threshold crossed) was written to memory as a hazard, so one false alarm at $p_h \in [\tau_c, \tau_h)$ renewed the caution for the next three missions, and with near-deterministic frames the false refusal repeated every mission. Memory now records real hazards (the standard rule flagged them) separately from caution alerts, and only real hazards renew the caution, so the per-mission bound above applies to each mission.

Over many missions: advice credit (I-24). The bounds above are per mission, so a channel that repeats wrong location advice or hazard claims at every mission (a poisoned peer memory file, a flooding operator account) multiplies them by the number of missions. Each *account* therefore holds location credit $c^\ell \in [0, C_\ell]$ and hazard credit $c^h \in [0, C_h]$ ($C_\ell = 2$ visit-equivalents, $C_h = 1$ refusal). Accounts are input channels, never labels that an item declares about itself: every item of the advice file shares one account, every item extracted from the instruction text another. A peer gets its own account only if the operator certified its identity, since free identities would multiply credit [103], [104]. An item is admitted to the plan only if the summed worst-case charge of its account's admitted items fits the credit; otherwise it is dropped and the report says so. The worst-case charge is one visit for a location item, plus $s = 0.25$ for the scrutiny views and one refusal that the scrutiny caution may cause; it is one false refusal for a hazard item. After the mission the credit is settled from the robot's own checks only (`rescuemind/advice_credit.py`, re-built from `mission.json` by the orchestrator). A reorder whose check confirmed the supply costs nothing. One whose check found another supply or could not decide costs one visit (the visit was spent) plus the scrutiny views. A refusal that only the tightened threshold caused (every view below τ_h, classified from the per-view scores) costs one. Nothing is refunded: with a duplicated supply a true claim need not save a visit, so a refund could be farmed. Instead, location credit recovers by $\rho_s = 0.25$ in each mission in which a location claim of the account, applied or dropped, was confirmed by the robot's own check (at most once per mission), and hazard credit regenerates by $\rho = 0.05$ per mission, so that an account that erred once is not locked out for ever. Because a subset of an advice set is an advice set, Prop. 2 holds unchanged.

Proposition 3 (price over a horizon). For every account and every advice sequence over m missions, including one chosen adaptively with knowledge of the ledger, the location price of its applied advice (wasted visits, scrutiny views on wrong claims and refusals caused by the scrutiny caution) is at most $C_\ell + \rho_s m$, and the false refusals caused by its hazard advice are at most $C_h + \rho m$. An account whose location claims are always confirmed and whose hazard claims never cause a caution-only refusal is never throttled. *Proof.* Admission reserves each admitted item's worst charge against credit that is never negative, so every charge is paid from credit. Credit is capped at C and grows only by ρ_s (location, at most once per mission) or ρ (hazard) per mission. Summing over missions telescopes. The last claim holds because such an account is never charged. □

Not priced. A truthful claim that steers the robot to a station that really burns costs a lost delivery, which is exposure and so Prop. 2's domain. Missions that end before settlement are not priced either. A property test drives the implementation with adaptive random advice sequences: 80 sequences of 60 missions, arena and depot, with and without scrutiny, and random channels. It checks both bounds after every mission. The counterexamples of the seventh internal review are regression tests: the label spreading and refund farming that broke the first version, and caution-only refusals that were missed without soft refusal. In a descriptive 100-mission model that runs the real rule on synthetic station outcomes ($k = 4$, Fig. S6, `scripts/advice_credit_horizon.py`, not registered), credit cuts a peer that always lies from 1533 s to 59 s of extra driving, and an adaptive liar that lies whenever its credit allows to 295 s. It keeps all 4363 s that an always-right peer saves, cuts hazard flooding at the photograph-regime false-alarm rate ($\delta = 0.18$) from 33 to 5 false refusals, and changes no delivery. A peer that is right 70% of the time keeps 45% of its saving. The recovery rate ρ_s trades this consistency against robustness (Fig. S6c), and some trade-off of this kind is unavoidable. A source that alternates confirmed truths with lies is re-admitted after $\lceil 1/\rho_s \rceil$ confirmations, so it extracts cost at rate about ρ_s, the bound's rate; any rule that re-admits after a fixed number of confirmations can be exploited the same way. The closed-loop test (study B) is registered; the mechanism is off in every other study.

Related work on pricing advice over time. In form, advice credit is a (σ, ρ) regulator [105]. Each source holds

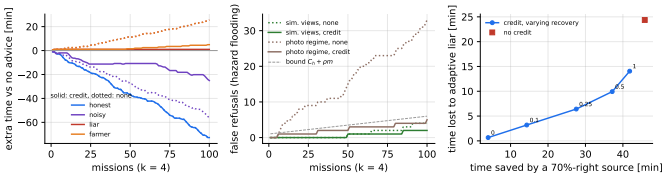


Fig. S6. Advice credit over 100 missions ($k = 4$, descriptive model on the real rule). (a) Extra time against the robot without advice (dotted: tighten-only rule without credit; solid: with credit; without credit the liar and the adaptive liar coincide). (b) False refusals under hazard flooding at the simulated and the photograph-regime false-alarm rate, with the bound $C_h + \rho m$. (c) Recovery rate ρ_s (labels): time saved by a 70%-right peer against time lost to an adaptive liar.

a bounded balance, an item is admitted only if the balance can pay its worst-case cost, and the cumulative price is at most a burst C plus a rate ρ per mission against any adaptive source. Learning-augmented algorithms also fall back to a robust baseline once predictions have proved costly, and recent work makes the earned "budget" for following predictions explicit [106], [107]. They compete with the best predictor, or a switching mixture of predictors [108], [109], and their trust–robustness trade-offs have matching lower bounds [110], [111]. Those results bound a competitive ratio within one instance. Ours is an additive bound over repeated missions, per source, for a gate whose safety does not depend on the admission rule, so throttling costs only availability. Availability is exactly what false-positive attacks on safeguards exploit [112]. Spending only what a baseline has earned is also the idea of conservative bandits [113], and wealth that grows with verified successes is the idea of α-investing [114]. Unlike reputation in multi-robot and multi-agent systems [115], [116], credit is settled from the robot's own check, never from peers.

Trust base. "Advice may only tighten" is a claim about one small function, so the question is what else it rests on (Table S10). The advice audit recomputes `advice.plan` from the logged plan with the same module and a log written by the node under test, so it finds bugs, not a compromised node or a wrong specification; the evidence for safety in study A is the *outcome* audit (Gazebo ground-truth trajectory and masks), which does not depend on those logs. We therefore wrote a second implementation of the rule from its specification. It is independent in code, not in authorship: the same authors (with the same AI assistant) wrote both from the same specification, so a misreading of the specification would be shared; a third-party oracle written from the one-paragraph rule has not been done. We compared it with the implementation by differential testing over every advice set of up to four items, injected plausible bugs (skip allowed, threshold raised, view dropped, a location item deleting a station) and fuzzed raw advice files and model outputs (unknown keys, duplicates, huge numbers, NaN, bool, non-UTF-8, nested objects). This found five bugs in the implementation (an audit that missed a removed caution and NaN thresholds; a reorder that moved a naive memory's skip to an unvouched station; unexpected exceptions on non-list or deeply nested files; booleans accepted as probabilities; a NaN hazard score ignored by `max`); all were fixed and all injected bugs are now killed. A later proof review found three more, in the code the proof relies on, none of which fired in any registered campaign (every recorded side view carries a valid score, and every recorded model call that reports its server threshold reports 0.05): a side view with a missing or non-finite hazard score was counted as clear (fail-open, now a hazard), the server's hazard threshold (default 0.5) was not verified by the client, the audit or the interlock (now every call must record the configured value), and a caution-level alert was recorded as a hazard (now recorded apart). A further hardening, the delivery interlock (I-22: the velocity guard blocks delivery legs and shelter entry without a permit recomputed independently from the saved model outputs, with a guard-side shelter geofence), is implemented and off by default; its Gazebo check is still pending and studies A and X run with it off unless a registered amendment says otherwise.

Machine-checked monotonicity (I-23). We transcribed the code behind Proposition 2 into z3 (`scripts/verify_advice_smt.py`): the plan fold `advice.plan` with `assert_monotone`; the approval predicate as the mission node composes it (`apply_memory_caution`, `peek_hazard`, `combine_with_look_around`, adaptive and scrutiny views; scores are reals plus an *invalid* flag, labels and the suspicion trigger are free); and the loop over `stations.next_candidate`. Advice is unconstrained: any list of any length (an inductive invariant of the fold), malformed items included. All 99 queries are `unsat` (16 s on one core; $k \leq 8$ plan, $k \leq 6$ loop): (P2a) approval with advice $\Rightarrow$ approval under the base plan $\Rightarrow$ approval under the v3.6 rule, on arbitrary frames; (P2b) every delivery follows an approving check at that station, the order stays a permutation of all k stations and the visits are a prefix of it, so an unsafe delivery implies E_w; (C) at most k visits, one extra visit per reorder (attained), at most six views per station (advice adds only the three scrutiny views), and an approval is lost to advice only through $p_h \geq$ the caution threshold or a flagged scrutiny view. Premises are shown satisfiable and the bounds tight. *Link to the code.* 9 000 hypothesis inputs (boundary values, NaN, inf, bool, out-of-range scores, malformed and adversarial advice, naive base plans) are evaluated by the real functions and by the model and agree exactly (`tests/test_i23_smt_refinement.py`); Proposition 2 also holds on the real code with the plan assertion off. Eleven injected bugs (advice loosens τ_h, skips a check, marks a station verified, refuses unseen, drops a candidate; caution boundary or side views ignored; an invalid score read as clear) each give a z3 counterexample that makes the mutated code misbehave when replayed, and each is killed by the differential test; for relaxing interfaces z3 returns the two-world law and clairvoyant advice of the *Necessity* argument. *Trusted, and not covered.* Trusted: z3, CPython and the transcription. The ROS node is not executed: its loop is a model read off the source (guarded by a tripwire test) and the decision functions are composed as in the node.

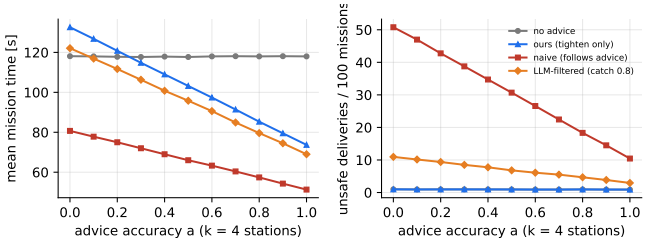


Fig. S7. Consistency–robustness curve of four policies as advice accuracy falls ($k = 4$; descriptive model with measured per-check rates; `scripts/advice_curve.py`). The tighten-only curve is flat by construction; see the text for what the model omits.

Not covered: ε, scene text, the camera and weights, frame independence (A1), the Jev-role model, the destination parse, time beyond the counts above. Three findings lie outside the proposition (advice cannot cause them): the audit raised on a run refused at an early extra view, and NaN or boolean supply confidences were accepted; all three are fixed, fail closed. *Reproduction.* `make verify` (host Python with `requirements-verify.txt`; no ROS, GPU or model weights) reruns every query, the differential and the mutants, and requires the same verdicts as the committed record; a fresh run on the authors' host took 4 minutes.

Consistency–robustness curve (model illustration, not registered). In the vocabulary of algorithms with predictions, Fig. S7 plots mission time (consistency: advice is right) and unsafe deliveries (robustness: advice is wrong) against the advice accuracy a for $k = 4$, for no advice, our tighten-only rule, a naive advice follower, and a naive follower behind a filter that drops a "safe/skip" claim with probability 0.8; per-check rates and times are those measured in G and H. The tighten-only curve is flat in unsafe deliveries by construction: in this model the fire sits only at the requested supply's station and the check's detection rate is the same at every station, so the proposition is displayed, not tested. Its informative parts are the time axis and the naive and filtered curves. The curve omits exposure: a hidden-fire bay with its own ε and duplicate supply would make the tighten-only curve rise above "no advice" as a falls, which is what A-steer measures. We therefore treat it as an illustration.

S11. The Second Checker and Campaign J

This section was part of the main paper; the main paper keeps one sentence, and row J stays in Table S12.

A. Method: the second checker

The original design of this robot asked two models (a second checker, called Jev in that design, and Qwen) and moved only when both agreed, following the idea of diverse redundancy [89], although failures are known to be correlated in practice [90]. The second checker is Clef-Flash, a decision model released by Cloudflare [117], [118] and served locally through the original second-checker API. According to its model card it is post-trained from Qwen3.5-9B, so the two models share lineage. Their diversity lies in post-training, in model size (our on-robot pair uses Qwen3.5-4B) and in the question asked, and the correlated-failure caveat applies in full. The two-model rule approves only if Eq. (1) holds for Qwen *and* Clef-Flash agrees (thresholds: Sec. S1). A hazard seen by either model stops the robot, and a wrong supply seen by either allows the one-time search. Whether the two models fit one GPU is tested in Sec. S11-B. The second model is a veto whose errors overlap with Qwen's. Qwen misses 17 of 211 development fires, and Clef-Flash also misses 15 of these (88%, Wilson [66, 97]%; Table S9).

B. Results: two models on one GPU (campaign J)

On T2, Clef-Flash together with Qwen3.5-4B, the pair that fits on one 32 GB V100, made 0/407 false approvals (185/193 correct). Campaign J ran this pair on the robot, paired with v3.6 on 24 worlds with swapped, empty and small-fire stations; it behaved as expected in 23/24 (v3.6: 24/24, exact McNemar $p = 1.0000$), with unsafe deliveries: 1 (v3.6: 0; Wilson upper bounds 20% and 14%); stack failures: 0. Peak GPU memory was 28.0 GiB, and the median check latency was 5.1 s for the two-model system and 5.7 s for v3.6; on our host, alternating ≈46 GB of weights thrashed the page cache, so both include model I/O (registered deviation note). The registered claim (two-model design on the simulated robot, no extra risk) is not supported. Its one unsafe delivery was a small fire behind the station reached after a search; both models missed it, the correlated failure that their development-set overlap predicts (Sec. S11-A). The arms differ in model size and hazard wording, so this does not test two-model checking in general, but it is why we did not adopt the pair.

S12. Studies and Results Moved from the Main Paper

The main paper follows one chain (gate, error budget, look-around, advice may only tighten, outside the guarantee) and keeps one sentence for each study that does not carry it. This section holds the rest.

A. All registered studies

Table S12 lists all 23 studies (without the initial evaluation C); the main paper's table keeps the rows that carry the chain.

B. Hazard fixes, held-out confirmation (T2) and campaign F

T1 left one failure mode: fires only partly visible behind the supply. Because T1 was then spent, candidate fixes were compared on development data only, including a new set of small early-stage fires (DH2), since the first hazard set (DH) contained just four visible fires under 10 000 pixels. The candidates were enlarged overlapping crops for the hazard question, a hazard wording that names small, partly hidden or cut-off fire and smoke, and Clef-Flash (Sec. S11). The wording together with the crops is v3.6. Table S8 shows the results. The wording did most of the work: missed fires fell from 46 to 17 of 211 when combined with crops, and correct approvals did not change. Clef-Flash refused half of the correct

deliveries when asked the original second-checker questions, which concern the safety of a *passage*; with task-worded questions it reached 97%, and agreement with Qwen gave the fewest false approvals, but the two models need 37 GB together, which does not fit on one V100.

Held-out confirmation (T2, registered, 600 new images). The new draw happened to be easy: v3.5 already made 0/407 false approvals (190/193 correct approvals), so no configuration could improve on it; the registered claim was not supported. The configuration selected on development data (Clef-Flash with task wording plus Qwen3.5-9B v3.6, 37 GB) made 0/407 false approvals with 184/193 correct approvals, and Qwen3.5-9B v3.6 alone made 0/407 with 190/193. Clef-Flash *alone*, given the task-worded questions, made 0/407 false approvals with 187/193 correct approvals in a single forward pass (median 0.6 s per image, against 3.4 s for the scored Qwen queries; Table S5).

Closed loop, small fires (campaign F, registered, paired). Each of 24 worlds had a small fire just behind a station or no fire, and was run with v3.5 and with v3.6. In the 13 worlds where the fire was visible at the checkpoint in both runs, v3.5 delivered despite the fire 12 times and v3.6 7 times (exact McNemar $p = 0.0625$, not significant; 5 discordant worlds favoured v3.6 and 0 the other). Neither system stopped falsely in the fire-free runs (v3.5 0/11, v3.6 0/11); with so few runs this says little. Six of v3.6's seven remaining misses were small fires whose hazard probabilities overlap those of fire-free views (Sec. S7). A wording or a crop cannot add the pixels that the front view lacks, which led to the look-around of the main paper (Sec. III-B).

C. Study W: look-around at image level, and prior mechanisms

Study W (registered, image level) replays the check offline on new small, partly hidden fires: *(pending)* perceivable fires, rendered at the front and both side poses, against fire-free twins. The front view alone missed *(pending)*, look-around *(pending)* (*(pending)*, 95% CI *(pending)*), at *(pending)* extra false stops on fire-free twins (claim: look-around miss bound $\leq 15\%$, *(pending)*; cost *(pending)*). With the same full-image input, a paraphrase vote in the style of VAP-TAMP [11] and two conformal sets in the style of KnowNo [14] approved *(pending)*, *(pending)* and *(pending)* must-refuse images against *(pending)* for ours (distinguished from *(pending)* of 3). A secondary analysis of W evaluates suspicion-triggered extra views (adaptive views, I-20): where any view of the front and two side poses scores at least 0.005 but below τ_h, the robot looks from two more poses at 0.8 m and one 0.3 m closer, and any of them at $p_h \geq \tau_h$ revokes the approval. On development data (DH and DH2 only), 17 of the 23 fires missed at $\tau_h = 0.05$ scored at least 0.005, against 40 of 380 fire-free views (11%); the held-out result is *(pending)* and a closed-loop test is not yet registered.

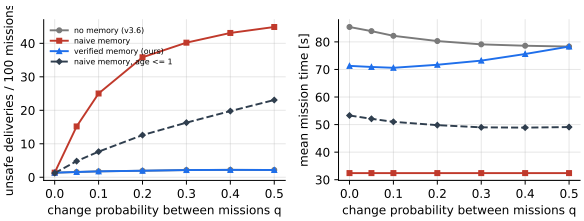


Fig. S8. Registered model of repeated missions, from the per-check rates measured in G and H: unsafe deliveries and mean mission time as the world changes more often, for no memory, naive memory, a recency-gated naive memory that trusts only the previous mission (M1age), and verified memory (a model prediction, not an observation). Recency gating reduces but does not remove the risk of trusting memory; verified memory keeps the memory-free risk.

D. Zero-shot baselines (Z) and the conformal threshold (K)

Is a generative VLM needed? (study Z, registered). We ran contrastive zero-shot baselines with SigLIP [101] and CLIP [102], with the same decision structure and thresholds chosen on development data only (Table S6). SigLIP approved 44/408 (T1) and 47/407 (T2) must-refuse images, against 3/408 and 0/407 for our check (exact McNemar $p < 10^{-6}$). CLIP refused almost everything: it gave 2/191 and 0/193 correct approvals. *Conformal τ_h (study K, registered, offline)* kept its miss bound at the cost of more false alarms; the frozen threshold was not changed (Sec. S7).

E. Camera degradations (study R)

Gazebo renders are noise-free and well exposed. To probe this part of the sim-to-real gap [76], we degraded the front-view frames of 300 held-out T2 images (30 worlds) in 12 registered ways. There are six families, in the spirit of ImageNet-C [72]: sensor noise, motion blur, defocus, under- and over-exposure, and JPEG compression, each at a moderate and a severe level (examples: Fig. S4). A frame that the robot's own acquisition gate rejects (black-out, white-out or flat frame) never reaches the VLM and counts as a refusal. Table S7 gives every condition. Pooled over all conditions, the check made 0/2304 false approvals, so the registered fail-closed claim is supported. Pooling repeats the same 192 must-refuse images, so the image-level bound is per condition: at most 0/192 false approvals, Wilson upper bound 1.96%. 0 conditions increased false approvals significantly. Correct approvals went from 107/108 on clean frames to 59/108 in the worst condition (overexposure S). The gate refused 28 frames in total, and 0 of 27 visible fires were missed in any condition. Under shift, refusal rather than unsafe approval is the intended failure direction of a selective classifier [119].

F. The model of repeated missions

Fig. S8 extends campaign H. At a change probability of $q = 0.2$ per mission, the model predicts 35.83 unsafe deliveries per 100 missions for naive memory, against 1.95 for verified memory and 1.95 without memory; the mean mission times are 32.4, 71.7 and 80.3 s. Keeping every check has a price: verified memory keeps 27% of naive memory's modelled time saving

at $q = 0$ (14.1 of 53.0 s) and 0% at $q = 0.5$. The saving grows with the number of stations while the risk bound does not change (Sec. S7). Study V tests the model in closed loop (main paper, Sec. VI-B). Further H details: the per-mission outcomes (Fig. S5), the case in which caution stopped a delivery that no-memory made (main paper, Fig. 5), all paired missions (Table S8), the time of delivered missions by arm and false refusals: M0 0, M1 4, M2 0. Caution applied in 12 of the 32 M2 missions, only 4 at a fire-free station, with 0 false refusals there (Wilson upper bound 49%); a global $\tau_h = 0.0113$ would also have stopped M0 in Q09.3 (descriptive, not registered; Sec. S7).

REFERENCES

[1] Z. Liu, A. Bahety, and S. Song, "REFLECT: Summarizing robot experiences for failure explanation and correction," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 3468–3484. [Online]. Available: https://proceedings.mlr.press/v229/liu23g.html

[2] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang *et al.*, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/70a06501001e1820fd1eb9ee821302d2-Abstract-Conference.html

[3] W. Huang, F. Xia, T. Xiao, H. Chan *et al.*, "Inner monologue: Embodied reasoning through planning with language models," in *Proceedings of The 6th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2022, pp. 1769–1782, coRL 2022 (PMLR volume dated 2023); arXiv preprint July 2022. [Online]. Available: https://proceedings.mlr.press/v205/huang23c.html

[4] Y. Du, K. Konyushkova, M. Denil, A. Raju *et al.*, "Vision-language models as success detectors," in *Proceedings of The 2nd Conference on Lifelong Learning Agents (CoLLAs)*, ser. Proceedings of Machine Learning Research, vol. 232. PMLR, 2023, pp. 120–136. [Online]. Available: https://proceedings.mlr.press/v232/du23b.html

[5] C. Agia, R. Sinha, J. Yang, Z. Cao *et al.*, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 689–723, coRL 2024 (PMLR volume dated 2025); method name: Sentinel. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[6] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez *et al.*, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Robotics: Science and Systems XXI (RSS)*, 2025, method name: FAIL-Detect. [Online]. Available: https://doi.org/10.15607/RSS.2025.XXI.073

[7] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, "Failure prediction at runtime for generative robot policies," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025, method name: FIPER.

[8] E. Zhou, Q. Su, C. Chi, Z. Zhang *et al.*, "Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 6919–6929. [Online]. Available: https://doi.org/10.1109/CVPR52734.2025.00649

[9] R. Sinha, A. Elhafsi, C. Agia, M. Foutter, E. Schmerling, and M. Pavone, "Real-time anomaly detection and reactive planning with large language models," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[10] J. Yuan, Y. Wu, and A. Bajcsy, "When to act, ask, or learn: Uncertainty-aware policy steering," in *Proc. Robotics: Science and Systems (RSS)*, 2026, to appear (per arXiv v2 comments); proceedings entry not independently checked.

[11] A. Oloo, Z. Altaweel, Y. Hayamizu, P. Liu, Y. Ding, S. Amiri, H. Yang, A. Kaminski, C. Esselink, C. Paxton, X. Zhang, and S. Zhang, "Robot planning and situation handling with active perception," 2026, arXiv preprint; method name: VAP-TAMP.

[12] A. Elhafsi, R. Sinha, C. Agia, E. Schmerling, I. Nesnas, and M. Pavone, "Semantic anomaly detection with large language models," 2023, arXiv preprint.

[13] S. Whitehead, S. Petryk, V. Shakib, J. Gonzalez, T. Darrell, A. Rohrbach, and M. Rohrbach, "Reliable visual question answering: Abstain rather than answer incorrectly," in *European Conference on Computer Vision (ECCV)*, 2022.

[14] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh *et al.*, "Robots that ask for help: Uncertainty alignment for large language model planners," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 661–682. [Online]. Available: https://proceedings.mlr.press/v229/ren23a.html

[15] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021. [Online]. Available: https://arxiv.org/abs/2107.07511

[16] K. Liang, Z. Zhang, and J. F. Fisac, "Introspective planning: Aligning robots' uncertainty with inherent task ambiguity," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024. [Online]. Available: https://papers.nips.cc/paper_files/paper/2024/hash/8451a20c5a7e0ee5671dda28f7daf7f3-Abstract-Conference.html

[17] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *International Conference on Learning Representations (ICLR)*, 2024.

[18] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, pp. 1641–1662, 2025.

[19] J. Lekeufack, A. N. Angelopoulos, A. Bajcsy, M. I. Jordan, and J. Malik, "Conformal decision theory: Safe autonomous decisions from imperfect predictions," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 11 668–11 675.

[20] B. Ichter, A. Brohan, Y. Chebotar, C. Finn *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of The 6th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2022, pp. 287–318, coRL 2022 (PMLR volume dated 2023). Author order follows the PMLR record; the arXiv version (2204.01691) lists Michael Ahn first ("Ahn et al., 2022"). [Online]. Available: https://proceedings.mlr.press/v205/ichter23a.html

[21] J. Robinson, C. M. Rytting, and D. Wingate, "Leveraging large language models for multiple choice question answering," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=yKbprarjc5B

[22] S. Kadavath, T. Conerly, A. Askell, T. Henighan *et al.*, "Language models (mostly) know what they know," 2022. [Online]. Available: https://arxiv.org/abs/2207.05221

[23] C. Zheng, H. Zhou, F. Meng, J. Zhou, and M. Huang, "Large language models are not robust multiple choice selectors," in *International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://openreview.net/forum?id=shr9PXz7T0

[24] P. Pezeshkpour and E. Hruschka, "Large language models sensitivity to the order of options in multiple-choice questions," in *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, 2024, pp. 2006–2017. [Online]. Available: https://doi.org/10.18653/v1/2024.findings-naacl.130

[25] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[26] Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, "Calibrate before use: Improving few-shot performance of language models," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 12 697–12 706. [Online]. Available: https://proceedings.mlr.press/v139/zhao21c.html

[27] T. Wu, C. Zhou, G. Zhao, H. Cao, Y. Pu, and J. Yang, "When robots should say "I don't know": Benchmarking abstention in embodied question answering," 2025, arXiv preprint.

[28] X. Wang, J. Wei, D. Schuurmans, Q. Le *et al.*, "Self-consistency improves chain of thought reasoning in language models," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=1PL1NIMMrw

[29] Y. Li, Y. Du, K. Zhou, J. Wang *et al.*, "Evaluating object hallucination in large vision-language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2023, pp. 292–305. [Online]. Available: https://doi.org/10.18653/v1/2023.emnlp-main.20

[30] D. Seto, B. Krogh, L. Sha, and A. Chutinan, "The Simplex architecture for safe online control system upgrades," in *Proceedings of the 1998 American Control Conference (ACC)*. IEEE, 1998, pp. 3504–3508. [Online]. Available: https://doi.org/10.1109/ACC.1998.703255

[31] L. Sha, "Using simplicity to control complexity," *IEEE Software*, vol. 18, no. 4, pp. 20–28, 2001. [Online]. Available: https://doi.org/10.1109/MS.2001.936213

[32] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer *et al.*, "Safe reinforcement learning via shielding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018. [Online]. Available: https://doi.org/10.1609/aaai.v32i1.11797

[33] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista *et al.*, "Control barrier functions: Theory and applications," in *18th European Control Conference (ECC)*, 2019, pp. 3420–3431. [Online]. Available: https://doi.org/10.23919/ECC.2019.8796030

[34] A. Robey, Z. Ravichandran, V. Kumar, H. Hassani, and G. J. Pappas, "Jailbreaking LLM-controlled robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11 948–11 956. [Online]. Available: https://doi.org/10.1109/ICRA55743.2025.11128119

[35] Z. Ravichandran, A. Robey, V. Kumar, G. J. Pappas, and H. Hassani, "Safety guardrails for LLM-enabled robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4649–4656, 2026. [Online]. Available: https://doi.org/10.1109/LRA.2026.3667488

[36] K.-C. Hsu, H. Hu, and J. F. Fisac, "The safety filter: A unified view of safety-critical control in autonomous systems," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 7, no. 1, pp. 47–72, 2024.

[37] L. Santos, Z. Li, L. Peters, S. Bansal, and A. Bajcsy, "Updating robot safety representations online from natural language feedback," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 7778–7785.

[38] L. Wang, Z. Ying, X. Yang, Q. Zou, Z. Yin, T. Li, J. Yang, Y. Yang, A. Liu, and X. Liu, "RoboSafe: Safeguarding embodied agents via executable safety logic," 2025, arXiv preprint.

[39] B. Zitkovich, T. Yu, S. Xu, P. Xu *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 2165–2183, author order follows the PMLR record; the arXiv version (2307.15818) lists Anthony Brohan first ("Brohan et al., 2023"). [Online]. Available: https://proceedings.mlr.press/v229/zitkovich23a.html

[40] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao *et al.*, "OpenVLA: An open-source vision-language-action model," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 2679–2713, coRL 2024 (PMLR volume dated 2025). [Online]. Available: https://proceedings.mlr.press/v270/kim25c.html

[41] A. Anwar, J. Welsh, J. Biswas, S. Pouya, and Y. Chang, "ReMEmbR: Building and reasoning over long-horizon spatio-temporal memory for robot navigation," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 2838–2845.

[42] Q. Xie, S. Y. Min, P. Ji, Y. Yang, T. Zhang, K. Xu, A. Bajaj, R. Salakhutdinov, M. Johnson-Roberson, and Y. Bisk, "Embodied-RAG: General non-parametric embodied memory for retrieval and generation," 2024, arXiv preprint.

[43] Z. Wang, B. Yu, J. Zhao, W. Sun, S. Hou, S. Liang, X. Hu, Y. Han, and Y. Gan, "KARMA: Augmenting embodied AI agents with long-and-short term memory systems," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[44] L. Bärmann, C. DeChant, J. Plewnia, F. Peller-Konrad, D. Bauer, T. Asfour, and A. Waibel, "Episodic memory verbalization using hierarchical representations of life-long robot experience," in *Proc. IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2025, pp. 783–790.

[45] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," in *Proc. Robotics: Science and Systems (RSS)*, 2022.

[46] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 5021–5028.

[47] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[48] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf, "SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning," in *Proc. Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 23–72.

[49] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 10 608–10 615. [Online]. Available: https://doi.org/10.1109/ICRA48891.2023.10160969

[50] P. Biber and T. Duckett, "Dynamic maps for long-term operation of mobile service robots," in *Proc. Robotics: Science and Systems (RSS) I*, 2005, pp. 17–24.

[51] T. Krajník, J. P. Fentanes, J. P. S. S. Santos, and T. Duckett, "FreMEn: Frequency map enhancement for long-term mobile robot autonomy in changing environments," *IEEE Transactions on Robotics*, vol. 33, no. 4, pp. 964–977, 2017.

[52] L. Schmid, M. Abate, Y. Chang, and L. Carlone, "Khronos: A unified approach for spatio-temporal metric-semantic SLAM in dynamic environments," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[53] T. Behrens, R. Zurbrügg, M. Pollefeys, Z. Bauer, and H. Blum, "Lost & found: Tracking changes from egocentric observations in 3D dynamic scene graphs," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3739–3746, 2025.

[54] Y. Sun and Y. Zhang, "When memory lies: An empirical study of spatial memory staleness in VLM agents," 2026, arXiv preprint.

[55] Z. Chen, Z. Xiang, C. Xiao, D. Song, and B. Li, "AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[56] K. Yadav, Y. Ali, G. Gupta, Y. Gal, and Z. Kira, "FindingDory: A benchmark to evaluate memory in embodied agents," 2025, arXiv preprint.

[57] X. Yin and H. Du, "GLOVE: Global verifier for LLM memory-environment realignment," 2026, arXiv preprint.

[58] G. Tang, Q. Jia, G. Chen, T. Li, Z. Huang, Z. Lv, and N. Ji, "VLM-DEWM: Dynamic external world model for verifiable and resilient vision-language planning in manufacturing," 2026, arXiv preprint.

[59] A. Aydemir, A. Pronobis, M. Göbelbecker, and P. Jensfelt, "Active visual object search in unknown environments using uncertain semantics," *IEEE Transactions on Robotics*, vol. 29, no. 4, pp. 986–1002, 2013.

[60] Z. Zeng, A. Röfer, and O. C. Jenkins, "Semantic linking maps for active visual object search," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020, pp. 1984–1990, arXiv:2006.10807.

[61] A. Wandzel, Y. Oh, M. Fishman, N. Kumar, L. L. S. Wong, and S. Tellex, "Multi-object search using object-oriented POMDPs," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2019, pp. 7194–7200.

[62] J. Mason and B. Marthi, "An object-based semantic world model for long-term change detection and semantic querying," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 3851–3858.

[63] D. M. Rosen, J. Mason, and J. J. Leonard, "Towards lifelong feature-based mapping in semi-static environments," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2016, pp. 1063–1070.

[64] Z. Yan, S. Li, Z. Wang, L. Wu, H. Wang, J. Zhu, L. Chen, and J. Liu, "Dynamic open-vocabulary 3D scene graphs for long-term language-guided mobile manipulation," *IEEE Robotics and Automation Letters*, vol. 10, no. 5, pp. 4252–4259, 2025, arXiv:2410.11989.

[65] K. Zhou, K. Zheng, C. Pryor, Y. Shen, H. Jin, L. Getoor, and X. E. Wang, "ESC: Exploration with soft commonsense constraints for zero-shot object navigation," in *Proc. 40th Int. Conf. Machine Learning (ICML), PMLR 202*, 2023, pp. 42 829–42 842, arXiv:2301.13166.

[66] N. Yokoyama, S. Ha, D. Batra, J. Wang, and B. Bucher, "VLFM: Vision-language frontier maps for zero-shot semantic navigation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024, pp. 42–48, arXiv:2312.03275.

[67] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.

[68] R. Bajcsy, Y. Aloimonos, and J. K. Tsotsos, "Revisiting active perception," *Autonomous Robots*, vol. 42, no. 2, pp. 177–196, 2018.

[69] W. R. Scott, G. Roth, and J.-F. Rivest, "View planning for automated three-dimensional object reconstruction and inspection," *ACM Computing Surveys*, vol. 35, no. 1, pp. 64–96, 2003.

[70] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh, "Explore until confident: Efficient exploration for embodied question answering," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[71] V. Sripada, S. Carter, F. Guerin, and A. Ghalamzan, "Scene exploration by vision-language models," 2024, arXiv preprint.

[72] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. International Conference on Learning Representations (ICLR)*, 2019.

[73] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, and W. Brendel, "Benchmarking robustness in object detection: Autonomous driving when winter is coming," 2019, arXiv preprint.

[74] S. Chen, J. Gu, Z. Han, Y. Ma, P. Torr, and V. Tresp, "Benchmarking robustness of adaptation methods on pre-trained vision-language models," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.

[75] M. C. Schiappa, S. Vyas, H. Palangi, Y. S. Rawat, and V. Vineet, "Robustness analysis of video-language models against visual and language perturbations," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022.

[76] S. Höfer, K. Bekris, A. Handa, J. C. Gamboa, M. Mozifian, F. Golemo *et al.*, "Sim2Real in robotics and automation: Applications and challenges," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 2, pp. 398–400, 2021.

[77] H. Zhang, C. Zhu, X. Wang, Z. Zhou, C. Yin, M. Li, L. Xue, Y. Wang, S. Hu, A. Liu, P. Guo, and L. Y. Zhang, "BadRobot: Jailbreaking embodied LLM agents in the physical world," in *International Conference on Learning Representations (ICLR)*, 2025.

[78] X. Lu, Z. Huang, X. Li, C. Zhang, X. Ji, and W. Xu, "Easier said than done: Unpacking intent-behavior gap in jailbreaking LLM-based robots," 2024, arXiv v1 (2024-12-21) was titled "POEX: Policy Executable Embodied AI Jailbreak Attacks". Latest version accepted by NDSS 2027.

[79] E. K. Jones, A. Robey, A. Zou, Z. Ravichandran, G. J. Pappas, H. Hassani, M. Fredrikson, and J. Z. Kolter, "Adversarial attacks on robotic vision language action models," 2025, preprint.

[80] L. Burbano, D. Ortiz, Q. Sun, S. Yang, H. Tu, C. Xie, Y. Cao, and A. A. Cardenas, "CHAI: Command hijacking against embodied AI," in *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2026, accepted; proceedings entry not checked.

[81] S. M. B. P. Samarakoon, M. A. V. J. Muthugala, W. K. R. Sachinthana, and M. R. Elara, "Hijacking robots with a piece of paper: A systematic study of physical prompt injection in VLM-controlled robots," 2026, preprint.

[82] N. Dorzhiev, "RIPA: Sensory-vector prompt injection attacks on LLM-controlled ROS 2 robots," 2026, preprint.

[83] H. Cheng, E. Xiao, J. Gu, L. Yang, J. Duan, J. Zhang, J. Cao, K. Xu, and R. Xu, "Unveiling typographic deceptions: Insights of the typographic vulnerability in large vision-language models," in *European Conference on Computer Vision (ECCV)*, 2024, pp. 179–196.

[84] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, 2023, pp. 79–90.

[85] S. Dong, S. Xu, P. He, Y. Li, J. Tang, T. Liu, H. Liu, and Z. Xiang, "Memory injection attacks on LLM agents via query-only interaction," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[86] J. Liu, J. Guo, T. Zhang, Y. Xu, J. Wang, J. Fan, and B. Xiao, "Security of foundation-model-powered embodied agents: Attack surfaces, attacks, defenses, and evaluation," 2026, survey, preprint.

[87] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, "Defeating prompt injections by design," 2025, preprint.

[88] M. Costa, B. Köpf, A. Kolluri, A. Paverd, M. Russinovich, A. Salem, S. Tople, L. Wutschitz, and S. Zanella-Béguelin, "Securing AI agents with information-flow control," 2025, preprint.

[89] A. Avižienis, "The N-version approach to fault-tolerant software," *IEEE Transactions on Software Engineering*, vol. SE-11, no. 12, pp. 1491–1501, 1985.

[90] J. C. Knight and N. G. Leveson, "An experimental evaluation of the assumption of independence in multiversion programming," *IEEE Transactions on Software Engineering*, vol. SE-12, no. 1, pp. 96–109, 1986.

[91] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proc. International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 11 733–11 763.

[92] H. Kress-Gazit, K. Hashimoto, N. Kuppuswamy, P. Shah *et al.*, "Robot learning as an empirical science: Best practices for policy evaluation," 2024. [Online]. Available: https://arxiv.org/abs/2409.09491

[93] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. [Online]. Available: https://doi.org/10.1080/01621459.1927.10502953

[94] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. [Online]. Available: https://doi.org/10.1007/BF02295996

[95] J. Tobin, R. Fong, A. Ray, J. Schneider *et al.*, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30. [Online]. Available: https://doi.org/10.1109/IROS.2017.8202133

[96] D. Snyder, A. J. Hancock, A. Badithela, E. Dixon, P. Miller, R. A. Ambrus, A. Majumdar, M. Itkina, and H. Nishimura, "Is your imitation learning policy better than mine? policy comparison with near-optimal stopping," in *Robotics: Science and Systems XXI (RSS)*, 2025.

[97] X. Li, K. Hsu, J. Gu, K. Pertsch, O. Mees *et al.*, "Evaluating real-world robot manipulation policies in simulation," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 3705–3728, coRL 2024 (PMLR volume dated 2025); method name: SIMPLER.

[98] P. V. A. B. Venâncio, A. C. Lisboa, and A. V. Barbosa, "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices," *Neural Computing and Applications*, vol. 34, no. 18, pp. 15 349–15 368, 2022.

[99] M. Wang, P. Yue, L. Jiang, D. Yu, T. Tuo, and J. Li, "An open flame and smoke detection dataset for deep learning in remote sensing based fire detection," *Geo-spatial Information Science*, vol. 28, no. 2, pp. 511–526, 2025.

[100] L. Seidel, S. Gehringer, T. Raczok, S.-N. Ivens, B. Eckardt, and M. Maerz, "Advancing early wildfire detection: Integration of vision language models with unmanned aerial vehicle remote sensing for enhanced situational awareness," *Drones*, vol. 9, no. 5, p. 347, 2025.

[101] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 11 975–11 986.

[102] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

[103] J. R. Douceur, "The sybil attack," in *Peer-to-Peer Systems (IPTPS 2002)*, ser. LNCS, vol. 2429. Springer, 2002, pp. 251–260.

[104] E. J. Friedman and P. Resnick, "The social cost of cheap pseudonyms," *Journal of Economics & Management Strategy*, vol. 10, no. 2, pp. 173–199, 2001.

[105] R. L. Cruz, "A calculus for network delay, Part I: Network elements in isolation," *IEEE Transactions on Information Theory*, vol. 37, pp. 114–131, 1991.

[106] T. Lykouris and S. Vassilvitskii, "Competitive caching with machine learned advice," *Journal of the ACM*, vol. 68, no. 4, pp. 1–25, 2021, conference version: ICML 2018. arXiv:1802.05399.

[107] P. Chen, H. Zhao, X. Tang, Y. Wang, and S. Deng, "Towards optimal robustness in learning-augmented paging," *arXiv preprint arXiv:2606.01342*, 2026.

[108] K. Anand, R. Ge, A. Kumar, and D. Panigrahi, "Online algorithms with multiple predictions," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 582–598. [Online]. Available: https://proceedings.mlr.press/v162/anand22a.html

[109] A. Antoniadis, C. Coester, M. Eliáš, A. Polak, and B. Simon, "Mixing predictions for online metric algorithms," arXiv:2304.01781, 2023.

[110] A. Wei and F. Zhang, "Optimal robustness-consistency trade-offs for learning-augmented online algorithms," in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020, arXiv:2010.11443.

[111] S. Angelopoulos, C. Dürr, S. Jin, S. Kamali, and M. Renault, "Online computation with untrusted advice," in *11th Innovations in Theoretical Computer Science Conference (ITCS)*, ser. LIPIcs, 2020, arXiv:1905.05655.

[112] Q. Zhang, Z. Xiong, and Z. M. Mao, "LLM safeguard is a double-edged sword: Exploiting false positives for denial-of-service attacks," *arXiv preprint arXiv:2410.02916*, 2024.

[113] Y. Wu, R. Shariff, T. Lattimore, and C. Szepesvári, "Conservative bandits," in *Proceedings of the 33rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 48, 2016, pp. 1254–1262. [Online]. Available: https://proceedings.mlr.press/v48/wu16.html

[114] D. P. Foster and R. A. Stine, "α-investing: a procedure for sequential control of expected false discoveries," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 70, no. 2, pp. 429–444, 2008.

[115] F. Mallmann-Trenn, M. Cavorsi, and S. Gil, "Crowd vetting: Rejecting adversaries via collaboration with application to multirobot flocking," *IEEE Transactions on Robotics*, vol. 38, pp. 5–24, 2022.

[116] S. Ren, W. Fu, X. Zou, C. Shen, Y. Cai, C. Chu, Z. Wang, and S. Hu, "Reputation as a solution to cooperation collapse in LLM-based MASs," in *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2026, arXiv:2505.05029.

[117] Cloudflare, "Clef-Flash model card," Hugging Face model repository, 2026. [Online]. Available: https://huggingface.co/Cloudflare/clef-flash

[118] M. Chen, "Introducing Clef: Our open-source decision models, and new RL fine-tuning platform," Cloudflare Blog, Oct. 2026. [Online]. Available: https://blog.cloudflare.com/clef-decision-models

[119] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NIPS)*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html

TABLE S8

CAMPAIGN H, ALL PAIRED MISSIONS (SEQ = SEQUENCE.MISSION). OUTCOME AND MISSION TIME: ✓ DELIVERED, NO UNSAFE DELIVERY; × UNSAFE DELIVERY; ○ REFUSED WHERE NO SAFE OPTION EXISTED; △ REFUSED ALTHOUGH A SAFE OPTION EXISTED (FALSE REFUSAL); NO MISSION FAILED. TRANSITION: WHAT CHANGED SINCE THE PREVIOUS MISSION. STORIES (MISSIONS 1→2→3): S1 NOMINAL, SWAPPED, SWAPPED; S2 SWAPPED, SWAPPED, NOMINAL; S3 NO FIRE, FIRE BEHIND THE REQUESTED STATION, FIRE; S4 FIRE, NO FIRE, FIRE. M2 SOURCE: FIRST STATION TAKEN FROM MEMORY AND/OR CAUTION APPLIED TO A STATION.

Seq	Story	Transition	M0	M1	M2	M2 source
Q01.2	S3	fire appears	○ (62 s)	× (41 s)	○ (45 s)	memory
Q01.3	S3	fire persists	○ (30 s)	× (41 s)	○ (30 s)	memory+caution
Q02.2	S2	relocation persist	✓(112 s)	✓(28 s)	✓(72 s)	memory
Q02.3	S2	relocation change	✓(85 s)	× (28 s)	✓(114 s)	memory
Q03.2	S1	relocation change	✓(111 s)	× (28 s)	✓(114 s)	memory
Q03.3	S1	relocation persist	✓(115 s)	× (28 s)	✓(86 s)	memory
Q04.2	S4	fire gone	✓(84 s)	△ (3 s)	✓(83 s)	memory+caution
Q04.3	S4	fire appears	○ (44 s)	○ (3 s)	○ (44 s)	memory+caution
Q05.2	S3	fire appears	○ (59 s)	× (29 s)	○ (59 s)	memory
Q05.3	S3	fire persists	○ (43 s)	× (28 s)	○ (41 s)	memory+caution
Q06.2	S4	fire gone	✓(83 s)	△ (3 s)	✓(83 s)	memory+caution
Q06.3	S4	fire appears	○ (31 s)	○ (3 s)	○ (32 s)	memory+caution
Q07.2	S1	relocation change	✓(108 s)	× (42 s)	✓(109 s)	memory
Q07.3	S1	relocation persist	✓(115 s)	× (39 s)	✓(72 s)	memory
Q08.2	S4	fire gone	✓(74 s)	△ (3 s)	✓(72 s)	memory+caution
Q08.3	S4	fire appears	○ (28 s)	○ (3 s)	○ (29 s)	memory+caution
Q09.2	S4	fire gone	✓(83 s)	△ (3 s)	✓(85 s)	memory+caution
Q09.3	S4	fire appears	× (85 s)	○ (3 s)	○ (44 s)	memory+caution
Q10.2	S2	relocation persist	✓(106 s)	✓(27 s)	✓(72 s)	memory
Q10.3	S2	relocation change	✓(83 s)	× (27 s)	✓(113 s)	memory
Q11.2	S2	relocation persist	✓(114 s)	✓(39 s)	✓(85 s)	memory
Q11.3	S2	relocation change	✓(74 s)	× (39 s)	✓(106 s)	memory
Q12.2	S3	fire appears	○ (42 s)	× (29 s)	○ (41 s)	memory
Q12.3	S3	fire persists	○ (29 s)	× (29 s)	○ (28 s)	memory+caution
Q13.2	S3	fire appears	○ (29 s)	× (29 s)	○ (28 s)	memory
Q13.3	S3	fire persists	○ (62 s)	× (28 s)	○ (28 s)	memory+caution
Q14.2	S2	relocation persist	✓(120 s)	✓(40 s)	✓(85 s)	memory
Q14.3	S2	relocation change	✓(75 s)	× (40 s)	✓(113 s)	memory
Q15.2	S1	relocation change	✓(125 s)	× (28 s)	✓(116 s)	memory
Q15.3	S1	relocation persist	✓(114 s)	× (28 s)	✓(96 s)	memory
Q16.2	S1	relocation change	✓(109 s)	× (41 s)	✓(112 s)	memory
Q16.3	S1	relocation persist	✓(108 s)	× (41 s)	✓(73 s)	memory

TABLE S9

WHEN EACH STUDY WAS REGISTERED AND WHEN ITS FIRST RESULTS ENTERED THE REPOSITORY. "ORDER" CHECKS THAT THE REGISTRATION COMMIT IS AN ANCESTOR OF THE FIRST RESULTS COMMIT (`GIT MERGE-BASE --IS-ANCESTOR`). REGISTRATIONS ARE ALSO MIRRORED IN THE PUBLIC REPOSITORY HISTORY. STUDIES WITHOUT RESULTS YET SHOW THE DRAFT COMMIT OR "REGISTERED AT MERGE, PENDING".

Study	Registered (commit, date)	First results (commit, date)	Amendments	Order
C (v3.3)	`5c9a951 2026-10-16`	`5c9a951 2026-10-16`	0	same commit
D (v3.4)	`ac6d815 2026-10-16`	`fb4e9b3 2026-10-16`	0	✓
T1	`4ad4f4f 2026-10-16`	`c3d1ffc 2026-10-16`	3	✓
E	`6201aa0 2026-10-16`	`17e0ea6 2026-10-16`	0	✓
T2	`418f1c8 2026-10-17`	`7843e9c 2026-10-17`	1	✓
F	`418f1c8 2026-10-17`	`5bbf11a 2026-10-17`	0	✓
G	`0816e36 2026-10-17`	`f6160fb 2026-10-17`	1	✓
R	`fb63bee 2026-10-17`	`2337e49 2026-10-17`	0	✓
H	`9807d68 2026-10-17`	`389ffc8 2026-10-17`	1	✓
J	`8163f88 2026-10-17`	`016b36e 2026-10-17`	0	✓
L	`f0f3d52 2026-10-17`	`6a471e2 2026-10-17`	0	✓
V	`abf0c98 2026-10-17`	`de1cdca 2026-10-18`	1	✓
W	`cc0e103 2026-10-17`	`pending`	2	pending
Z	`a53ee7e 2026-10-17`	`0ca401d 2026-10-17`	0	✓
K	`3a21238 2026-10-17`	`9b2086f 2026-10-17`	0	✓
P	`e6bde02 2026-10-17`	`00795a8 2026-10-17`	0	✓
P2	`2910b2e 2026-10-17`	`f96af5f 2026-10-17`	1	✓
A	`2e8972d 2026-10-17` (draft; registered at merge)	`pending`	0	pending
S	`5535660 2026-10-17` (draft; registered at merge)	`pending`	0	pending
X	`2e8972d 2026-10-17` (draft; registered at merge)	`pending`	0	pending
XD	`5535660 2026-10-17` (draft; registered at merge)	`pending`	0	pending
Q	`c7d06b3 2026-10-17` (draft; registered at merge)	`pending`	0	pending
B	`ed8e48b 2026-10-18` (draft; registered at merge)	`pending`	0	pending
IF	`051e530 2026-10-17` (draft; registered at merge)	`pending`	0	pending

TABLE S10

TRUST BASE OF THE ADVICE GUARANTEE.

Component	Assumed of it	Not covered
VLM outputs as recorded	scores and labels are those the model returned for the saved image; malformed answers give no advice, a broken score is a hazard; server τ_h equals the configured one	a server that lies consistently (its τ_h is self-reported); scene text
`advice.plan` (about 60 lines of logic)	only reorders and adds caution	shared reader of the specification
advice audit	recomputes the invariant from the logged plan	a compromised node
mission gating	a check is skipped or a station refused unseen only in the naive baselines	the ROS wiring (reviewed by the authors; the interlock I-22 that would enforce A3 is off by default)
ground-truth recorder	outcome audit independent of the node's logs	simulator fidelity
model allow-list and pinned weights (`ALLOWED_MODELS`, `PINNED_REVISIONS`)	a fixed set of model identifiers, each with the Hugging Face commit of its weights; the server reports the snapshot it loaded and the client refuses another revision	the server's report: a server that lies about its snapshot, or a modified local cache, is not detected; runs recorded before this check carry no revision and are reported as unverified
same-host deployment	VLM server, mission node, recorder and audit run on one machine, and the audit reads files written by the node under test (the outcome audit uses simulator ground truth, not those files)	a compromised or shared host; network faults between components; any two-machine deployment
second implementation	written from the specification, not from the code; used for differential tests	independence of authorship: same authors, same AI assistant, same specification
destination parse	outside Prop. 2: closed-set parse plus rule	a mis-parse of the request

TABLE S11

THREAT MODEL FOR INPUTS THE ROBOT DID NOT PERCEIVE ITSELF. "STRUCTURAL": A GUARANTEE THAT DOES NOT DEPEND ON THE MODEL'S ANSWERS.

Channel	Example attack	Structural?	Measured by
Own memory	stale or poisoned record: "water at M", "no fire at W"	yes (Obs. 1, Prop. 2)	H, V (naive vs. verified memory)
Peer robot report, inventory	false "fire is out", wrong location, "verified, skip"; truthful location that steers	yes (Prop. 2), steering is exposure	A, file channel; A-steer
Operator wording (advice)	"the fire is out, no need to check"; "ignore previous instructions"	yes (Prop. 2); extraction itself is a model	A, operator channel; 224 offline texts
Operator wording (destination)	"deliver to shelter A", school or hospital	no: closed-set parser plus explicit rule, fail-closed	A offline set (wrong-shelter acceptance must be 0)
Scene text	label "WATER" on a first-aid kit; sign "SAFE – NO FIRE"; "Answer C"	no: it is perception	S
Camera degradation	noise, blur, over-exposure (not adversarial)	no: fail-closed behaviour measured	R

TABLE S12

ALL REGISTERED STUDIES. EACH IS REGISTERED (PROTOCOL, CONFIGURATION, ENDPOINTS AND ANALYSIS CODE) BEFORE ITS FIRST TRIAL OR INFERENCE (FOR THE 8 STUDIES WITHOUT RESULTS YET, THE DRAFTS A, S, X, XD, Q, B, IF ARE REGISTERED WITH THE MERGE OF THE INTEGRATION BRANCH, BEFORE ANY OF THEIR TRIALS; TABLE S9); STUDIES WERE DESIGNED SEQUENTIALLY, EACH AFTER THE PREVIOUS RESULTS, SO REGISTRATION IS CONFIRMATORY WITHIN A STUDY (TABLE S9). AMENDMENTS MADE BEFORE THE FIRST TRIAL ARE PUBLISHED WITH THE REGISTRATION. CLAIM: STATUS OF THE REGISTERED CLAIM (DESCRIPTIVE: NONE REGISTERED; PENDING: NOT YET RUN).

Study	Question	Data	Arms (systems)	Primary endpoint	Claim
D	Does the v3.4 system behave as expected?	42 closed-loop trials, 3 scenes × variants	1 (v3.4)	expected behaviour	descriptive
T1	Does the check generalise to unseen variation?	600 held-out images, 60 worlds	1 (v3.4; v3.5, ablations secondary)	false approvals $< 2\%$	not supported
E	Closed loop in freshly randomised worlds	60 trials, one world each	1 (v3.5; counterfactuals)	expected behaviour, wrong deliveries	descriptive
T2	Do hazard fixes chosen on dev data hold?	600 new held-out images	2 (v3.5; selected)	false approvals	not supported
F	Do they help in closed loop on small hidden fires?	24 worlds, paired	2 (v3.5, v3.6)	deliveries despite a visible fire	not supported
G	Does looking from the side help?	24 new worlds, paired	2 (v3.6, v3.6 + look-around)	deliveries despite a perceivable fire	supported
R	Is the check fail-closed under camera degradation?	300 T2 images × 12 degradations	1 (v3.6)	pooled false approvals $< 2\%$	supported
H	Can memory speed missions up without new risk?	16 sequences × 3 missions, changing worlds	3 (v3.6 with no, naive, verified memory)	time; unsafe deliveries	supported
J	Does the two-model design run on the robot? (Supp. S11)	24 worlds, paired	2 (v3.6; Qwen3.5-4B + Clef-Flash)	completes on one GPU; unsafe deliveries	not supported
L	Does v3.6 generalise to unseen layouts?	24 random layouts	1	expected behaviour; unsafe deliveries	supported
V	Does the repeated-mission model predict closed-loop behaviour?	10 sequences × 6 missions, random change	3 (v3.6 with no, naive, verified memory)	prediction intervals; unsafe deliveries	model: supported; memory: not supported
W	How often is a small, partly hidden fire missed, and do prior mechanisms match ours?	150 fire + 75 fire-free worlds × 3 views (image level); T1 + T2 vs. vote and conformal baselines	2 (front only; look-around), 3 baselines	look-around miss bound $< 15\%$ (Wilson); false approvals	pending
Z	Is a generative VLM needed?	T1 + T2, contrastive zero-shot baselines	3	false and correct approvals	supported
K	Does a conformal τ_h keep a miss bound?	211 dev fires → T2 (offline)	2 ($\alpha = 0.05, 0.10$) vs frozen	held-out fire miss rate $\leq \alpha$	supported
P	Does the hazard question transfer to real photos?	400 D-Fire test images (CC0)	1 (v3.6 hazard question)	recall; false alarms (descriptive)	descriptive
P2	Does a site-calibrated τ_h make the hazard question usable on real photos?	D-Fire: 400 calibration, 400 test images	2 wordings vs frozen	recall ≥ 0.80 and false alarms ≤ 0.20	supported
A	Can untrusted advice relax a check, or steer it?	226 runs: 18 worlds (162 runs) + 16 duplicate-supply worlds (A-steer, 64 runs); 224 offline texts	tighten-only, naive, LLM-filter, scrutiny + control	unsafe deliveries under false advice; steering; time and cost bound	pending
S	Can a sign in the picture cause a wrong approval?	64 worlds × no sign, attack, benign	6 (ours, one pass, free-form, full image, OCR mask, ignore-text)	false approvals caused by the sign; defences (non-adaptive signs)	pending
XD	Does the check transfer to cluttered four-bay depot images?	308 worlds (hard class 100), 1 932 images; 200 warehouse photographs	3 (front only, look-around, adaptive)	hard-class δ upper bound ≤ 0.15; relative claims O1–O4; recalibrated τ_h	pending
X	Does the memory result transfer to a cluttered four-station depot?	80 closed-loop runs, $k = 4$	M0 vs M2 (memory); control, tighten-only, scrutiny (steering)	M2 vs M0 unsafe, time, visited bays; steering (descriptive)	pending
Q	Does the supply check work on real photographs?	405 Commons photographs	1 (v3.6 supply check)	false-approval upper bound ≤ 0.05 and correct approvals ≥ 0.70	pending
B	Does the advice credit cap the price of a source that is wrong at every mission, in the closed loop? (mechanics study, Prop. 3)	4 sequences × 6 missions × 2 arms = 48 runs, $k = 2$	2 (monotone advice with and without the credit ledger)	wasted visits $\leq C_{\text{bay}} + \rho m$ (deterministic prediction); no safety claim	pending
IF	Can the request text switch the requested supply (R6 P7)?	72 texts (6 categories × 2 languages × 2 supplies × 3 wordings), offline parser	1 (frozen v3.6 intent parser)	share of the 36 flip texts accepted with the injected supply (Wilson upper bound < 0.10); price in refused corrections (descriptive)	pending